

Age Set versus Kin: Culture and Financial Ties in East Africa[†]

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We study how social organization shapes patterns of economic interaction and the effects of national policy, focusing on the distinction between age-based and kin-based groups in sub-Saharan Africa. Motivated by ethnographic accounts suggesting that this distinction affects redistribution, we analyze a cash transfer program in Kenya and find that in age-based societies there are consumption spillovers within the age cohort, but not the extended family, while in kin-based societies we find the opposite. Next, we document that social structure shapes the impact of policy by showing that Uganda's pension program had positive effects on child nutrition only in kin-based societies. (JEL H23, I12, I38, J13, O15, Z13)

Informal income transfers are pervasive in low-income contexts. They influence people's everyday lives, making it possible to ward off periodic hardship, and the pattern and scale of these transfers shape the impact of public policy. In economic analysis, families or villages are the social units in which informal economic interactions are typically assumed to take place and in which individuals form financial links in order to smooth consumption in response to changing economic conditions (Townsend 1994, 1995).¹ Anthropologists since Boas (1887); Lowie (1917); and Benedict (1934), however, have documented vast and persistent differences in the pattern of social and economic exchange across cultures and argue that even seemingly clear-cut social groupings like the family vary widely in relevance. These differences in the dominant social organization lead to widespread variation in traditional patterns of allegiance, which persist and do not always adapt to changing

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¹There is also a growing realization of the value of direct network measurement for understanding financial transfers and analyzing policy (see Breza et al. 2019, for a review); many contexts pose limits on such detailed data collection, however, leaving us with a limited understanding of global variation in the structure of financial ties.

individual or regional economic conditions. In this framework, the relative importance of different social structures, including those outside the family, determines the pattern and strength of financial ties and, as a result, may shape the spillover effects of development policy.

This paper empirically investigates the extent to which long-standing features of social organization shape present-day financial ties and predict the distributional effects of public policy. Our focus is the stark distinction between societies organized around kin-based social groups and societies organized around age-based social groups. In kin-based societies, the main social group is the extended family and it is within the kin group that social allegiance and financial ties are often strongest (La Ferrara 2007; Alesina and Giuliano 2010, 2014; Kinnan and Townsend 2012; Lowes 2018a; Bau 2021). In age-based societies (or “age set” societies), however, the primary social unit is the age set, a group of individuals who are of similar age and initiated into adulthood at the same time, and the strongest feelings of obligation are between individuals of the same age set. While uncommon in the West, age set organization exists around the world and is particularly prevalent in Africa, where we estimate over 200 million people belong to ethnic groups in which age sets are the traditionally dominant form of social organization.² As a result, age set organization is described by Lowie (1921) as the most important example of an alternative to kinship-based organization (Ritter 1980). Nevertheless, the impact of age-based organization on economic activity remains largely unknown.

Age set organization has been thoroughly studied by anthropologists and their qualitative accounts suggest that in age set societies, within-cohort economic ties are substantially stronger than within-family and intergenerational ties (Radcliffe-Brown 1929; Dyson-Hudson 1963). According to Eisenstadt (1954, p. 101), “[R]elations between age mates [involve] general and permanent obligations of cooperation, solidarity, [and] help, closely resembling, in this respect, the family and kinship.” Still today, in age set societies in Kenya it is “toward age mates that men look for support rather than brothers or close patrilineal kin” (Spencer 2014, p. 48). Age mates are initiated together at a young age and pass through each phase of life together, creating a strong and life-long bond. Across studied examples, age sets lead to a strikingly different structure of allegiance and pattern of economic ties, even compared to kin-based groups living in the same country or region (Morton 1979).

We provide direct evidence of the impact of age-based versus kin-based organization on the strength of financial ties, focusing on east Africa, a setting in which these two prominent forms of social organization exist in the same areas. Our main hypothesis is that age set organization encourages stronger within-cohort links and weaker within-family and intergenerational links. This distinction leads to markedly different, yet predictable, spillover effects of public policy since resources are more likely to be distributed within the age cohort but less likely to be distributed within the kin group or across generations within the family. In order to investigate these

²Online Appendix A.6 describes the construction of our population estimate, which we compute by linking an extensive set of ethnographic accounts of social structure with modern ethnicity-level population data. Outside of Africa, age set organization has been studied in Brazil (Welch 2010) and the Pacific Islands (Claessen and Skalník 1978).

hypotheses, we compile data on the social structure of ethnic groups in Kenya and Uganda, the focus of our analysis. Information on the presence of age set organization is not available from standard ethnographic databases, including Murdock's (1967) *Ethnographic Atlas*. Therefore, we use a wide range of sources to determine whether age sets are prevalent in each society (see online Appendix A.1).³

The first part of our empirical analysis focuses on a cash transfer experiment in Kenya in order to directly investigate how exogenous income changes are redistributed in age set societies, compared to kin-based societies. In particular, we study a randomized evaluation of Kenya's Hunger Safety Net Program (HSNP), a large-scale cash transfer program that features both within-sample variation in age set organization and all data collection required to investigate our hypotheses.⁴ In the study, which took place across 48 sublocations in northern Kenya, households were determined eligible to receive the cash transfers via one of several targeting mechanisms; one-half of the sublocations were assigned to the treatment group, where delivery of the transfers began immediately, while the second half were assigned to the control group, where transfer delivery was delayed by two years.⁵ Crucially, members of age-based and kin-based societies are in both the eligible and noneligible populations, in both treated and control sublocations.

We first exploit the cash transfer experiment and a triple-difference empirical design to estimate within-cohort consumption spillovers, separately for members of age-based and kin-based societies. Comparing consumption patterns of *transfer-ineligible* households in treatment and control sublocations, we find that randomly increasing the income of an individual's cohort members increases consumption spending in age set societies and has no effect in kin-based societies. On average, every treated cohort member in an age set society increases consumption spending for members of the spillover group by US\$23, corresponding to 13 percent of the size of the transfer itself and 8 percent of median local income. As a falsification exercise, we document that there are no estimated spillover effects in age set societies from transfers to members of other cohorts in the same village, consistent with the effect being driven by age set organization itself.

These initial findings suggest that age set societies have large within-cohort consumption spillovers that are absent in kin-based societies; however, age set ties could be either *in addition to* family ties or, as suggested by some ethnographic accounts, *instead of* family ties. The HSNP evaluation did not directly collect data on extended family members or kin networks that could be used to estimate within-family spillover effects; therefore, we use two alternative strategies. First, while we find consumption spillover effects within sub-clans in kin-based societies, we show that

³We focus on the full set of ethnic groups listed for each country in the latest round of the *Demographic and Health Surveys*. We were able to definitively determine the dominant form of social organization for 99 percent of the studied sample in Kenya and 85 percent of the sample in Uganda, where we find members of age set societies comprise 51 percent and 31 percent of the population, respectively. Section I and online Appendix A.1 describe this process in detail.

⁴As will become clear, our analysis requires local variation in age set organization (in both treatment and control groups), as well as information on household-level ethnicity, age composition, and consumption. The HSNP evaluation data are public and available at <https://microdata.worldbank.org>. The original analysis of the program evaluation data was conducted by Mertens et al. (2013).

⁵Sublocations are super-sets of villages and are a standard and economically important administrative unit in Kenya.

these effects are entirely absent in age set societies.⁶ Second, to more directly investigate intergenerational ties within the family, we exploit the fact that one of the experiment's treatment arms was designed to simulate a pension program. It is often hypothesized that pension programs benefit children because grandparents invest in their grandchildren (Bertrand, Mullainathan, and Miller 2003; Duflo 2003). However, if intergenerational ties are weak in age set societies, pension recipients may be less likely to invest in children. Consistent with this conjecture, we find that pension grants led to substantial increases in educational investment in kin-based societies, while these effects are entirely absent in age set societies.⁷

Together, this first set of findings suggests that variation in social structure generates distinct patterns of financial ties. These differences are predictable using ethnographic descriptions of social organization, suggesting that ethnographic work could be a useful tool for approximating local network structures. Next, we directly investigate how local variation in social structure shapes the impact of national development policy.

The second part of our empirical analysis studies the staggered introduction of Uganda's national Senior Citizen Grant (SCG), a countrywide pension program for individuals over 65, and how variation in social organization affects the consequences of this large-scale policy. The Ugandan government anticipated that the program would have broad positive effects on *children*, which was a key goal of the program. According to the website of Uganda's Ministry of Gender, Labour and Social Development, the agency in charge of the SCG:

Although the SCGs target older people, the grants benefit more than only the senior citizen beneficiaries. [They also] have significant impact on development outcomes as older people tend to invest a portion of their grant money in meeting their grandchildren's nutritional, health, and education needs.⁸

To investigate the impact of the SCG on investment in children, we exploit the fact that the SCG was piloted in 2011 in 15 districts, combined with the fact that both pilot and non-pilot districts contain members of age set and kin-based societies. We use the 2016 round of Uganda's Demographic and Health Survey (DHS) to measure the *potential pension exposure* of each extended household using information on the age distribution of its members, as well as detailed anthropometric information about each child. Using a triple-difference identification strategy, analogous to our analysis of the HSNP, we find a large positive effect of the pension program on several measures of child nutrition in kin-based societies and no observable effect on any measure of child nutrition in age-based societies. The effect sizes are large relative to in-sample variation in child health: an additional year of the program increases child weight by 0.15 standard deviations and reduces the likelihood of

⁶ Sub-clans are a proxy for the extended family, but sub-clan names are self-reported in our data and we do not observe genealogical relationships among members or know whether all groups use the same definition for clan membership.

⁷ This result is not driven by lower economic engagement or altruism of older members of age set societies in general, since we find large within-cohort consumption spillovers after restricting the sample to older individuals.

⁸ See: <https://socialprotection.go.ug/wp-content/uploads/2018/07/Disability-response-in-the-Senior-Citizens-Grant-1.pdf>.

malnourishment by 5.5 percent in kin-based societies, and has no effect on any parameterization of child weight in age set societies.

We report several additional results that support a causal interpretation of the findings. First, we construct a placebo pension exposure measure based on extended household members who are *just below* the age of pension receipt. We find no differential effect on societies with and without age sets of this placebo measure, suggesting that the effect is not driven by the selection of households with more or fewer older household members. Second, we replicate our baseline estimates using the 2006 *round of the DHS*, prior to the introduction of the pension program pilot. Again, we find no differential effects, indicating that the results are not driven by preexisting trends. Third, we find qualitatively similar results using an alternative estimation strategy that only exploits program variation across districts and does not exploit the additional variation in program exposure across households. These estimates indicate that the baseline results are not driven by differential changes in pension program exposure, or any other feature of household composition, in response to the pension pilot program.

Additional evidence presented throughout the analysis shows that the different structure of financial ties and policy spillovers in age set societies is not driven by other cultural or social characteristics. First, we present a broad range of qualitative, ethnographic work on age set organization in Section I that is consistent with our findings and proposed mechanism. Second, both across ethnicities in our sample and across households in each empirical setting, we do not find evidence that age set organization is strongly correlated with other observable features of culture or social organization, including customs related to marriage, inheritance, gender, and local politics; household composition; baseline measures of child nutrition; the prevalence of violent conflict; and precolonial economic development. This could be due to the fact that our relatively narrow geographic focus holds many features of society fixed. While social structure is not randomly assigned and likely co-evolves with other features of society, these results indicate that our findings are not simply driven by preexisting trends or large differences between age set and kin-based societies in characteristics that might generate the results. Finally, controlling directly for the role of ethnicity-level political and economic organization does not affect our estimates in either part of the empirical analysis.

We conclude with a discussion of the broader implications of our findings for understanding economic inequality and vulnerability. Our results indicate that in kin-based societies, where we find strong connections between members of different ages, individuals may be well positioned to smooth consumption over the life-cycle. In age set societies, however, where these ties are weak, individuals at stages of life with less income earning potential (e.g., the young and the old) might consume systematically less and have fewer resources at their disposal. Returning to the expenditure data from the HSNP survey, we indeed find that while consumption is relatively flat over the life cycle in kin-based societies, it is hump-shaped in age set societies, suggesting that age set systems may generate greater inequality across age cohorts and leave the young and old particularly at risk. In kin-based societies, however, we estimate greater inequality across sub-clans and find that the poorest sub-clans in kin-based societies consume substantially less than the poorest sub-clans in age set societies. By cutting across

families, age sets may prevent any extended family from being left without access to resources or any one clan from accumulating much more than another. These empirical patterns are a final indication that social organization is an important determinant of the structure of financial ties, and that a better understanding of local culture could guide policy targeting and design.

Our findings build on a range of work investigating the economic impacts of social structure. Most prior work has focused on features of kinship structure, including whether inheritance is patrilineal or matrilineal (La Ferrara 2007; La Ferrara and Milazzo 2017; Lowes 2018a, b), whether the society is organized into segmentary lineages (Moscona, Nunn, and Robinson 2017, 2020), whether residence is matriloccal or patrilocal (Bau 2021), and whether family ties are strong or weak (Greif 1994; Alesina and Giuliano 2010, 2014; Akbari, Bahrami-Rad, and Kimbrough 2017; Enke 2019; Schulz et al. 2019; Schulz 2020; Ghosh, Hwang, and Squires 2021). Our results highlight the importance of social organizations other than the family and how strong “horizontal ties” lead to distinct patterns of economic interaction.

The second part of our analysis builds especially on existing work probing the relationship between traditional customs and intergenerational transfers (Duflo 2003; Anderson 2007; La Ferrara and Milazzo 2017; Jayachandran and Pande 2017; Corno, Hildebrandt, and Voena 2020; Ashraf et al. 2020; Bau 2021). The distributional effects of any development program can hinge on the strength of intergenerational investments, which we find are substantially curtailed in age set societies. Our results also build on a small body of work documenting that cultural context shapes the impact of policy (Ashraf et al. 2020); while there is growing awareness that the effects of any particular development policy could vary widely across cultures (World Bank 2015), empirical analysis at this interface is limited.

Our study also contributes to a large literature investigating informal risk sharing and redistribution (Rosenzweig and Stark 1989; Townsend 1994, 1995; Udry 1994; Ravallion and Chaudhuri 1997; Fafchamps 2003, 2011; Fafchamps and Gubert 2007).⁹ This paper is especially related to a growing body of work on the relationship between risk sharing and the structure of local networks (see Breza et al. 2019, for a review), including evaluations of risk pooling within family and caste groups (Munshi and Rosenzweig 2006; Kinnan and Townsend 2012; Ambrus, Mobius, and Adam Szeidl 2014; Angelucci, De Giorgi, and Rasul 2018).¹⁰ Remaining agnostic about the efficiency of each network structure, we show that differences in social organization lead to distinct redistribution patterns, even among close ethnic groups subject to very similar economic conditions. An implication of this finding is that ethnographic data could be an important tool for predicting network structure where direct measurement is impossible.

Last, our findings contribute to an established anthropological literature on the social role of age set organization and other forms of “cross cutting ties”—relationships that cut across family-based groups (see Bernardi 1985). Ethnographic accounts suggest that age set societies have a distinct pattern of social and economic

⁹A range of theoretical work examines the formation of financial networks, mostly focused on the desire to minimize consumption risk (Kocherlakota 1996; Bramoullé and Kranton 2007; Bloch, Genicot, and Ray 2008; Ambrus and Elliott 2020).

¹⁰See also, Morduch (2005), Mazzocco and Saini (2012), and Mobarak and Rosenzweig (2013) on caste networks in India, as well as Angelucci and De Giorgi (2009) on the importance of family ties.

interactions, in which within-cohort ties are favored over family ties (Lowie 1921; Radcliffe-Brown 1929; Eisenstadt 1954; Dyson-Hudson 1963; Bohannan 1964; Morton 1979; Kurimoto and Simonse 1998; Spencer 2014). While most ethnographic studies are from several decades ago, we show empirically that age set organization shapes modern financial ties and the impact of public policy.

This paper is organized as follows. The next section describes age set organization, as well as our sample, and discusses qualitative work on the affect of age sets on financial ties. Section II investigates consumption spillover effects in age-based compared to kin-based societies in a large unconditional cash transfer experiment in Kenya. Section III analyzes how kin-based versus age-based social structure shapes the effects of Uganda's national pension program on children. Section IV discusses additional implications of our findings and Section V concludes.

I. Age Set Systems

A. Defining and Identifying Age Set Organization

Age set societies are ethnic groups in which the dominant social group is the “age set,” a group of men, and sometimes women, of a similar age. Individuals in an age set are “initiated” into adulthood at the same time (and often under a common name), move through life together, and turn to each other for social and economic support. In the words of Radcliffe-Brown (1929, p. 21), one of the first to develop the terminology, an age set is defined as:

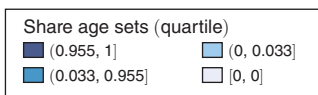
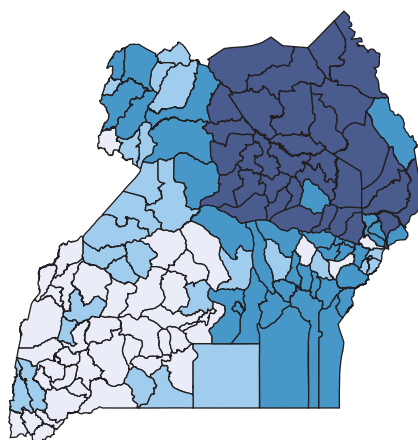
A recognized [group] consisting of persons (often male persons only) who are the same age [...] In Africa, at any rate in east and south Africa, an age set is normally formed of all those males who are initiated at one time. Once a person enters an age set [...] he remains a member of the same age set for the remainder of his life.

These age-based groups often take priority over family or kin relationships, and individuals have a stronger sense of obligation to their age set than to their extended family (Eisenstadt 1954).

Age set societies are widespread, and they are characterized by Lowie (1921) as one of the most prominent examples. Examples of societies with prominent age sets include the Oromo of Ethiopia (population: 27 million); the Zulu and Sotho of South Africa (population: 11 and 6.5 million, respectively); the Maasai and Kikuyu of Kenya and Tanzania (population: 1.65 and 8.1 million, respectively); the Iteso and Karimojong of Uganda (population: 3.2 and 1.2 million, respectively); and the Tiv of Nigeria (population: 6.5 million).¹¹ Based on our data, there are roughly 210 million individuals in sub-Saharan Africa today who are members of societies with a prominent age set system (see online Appendix A.6).

¹¹For references on the age set organization of the aforementioned groups, see Baxter et al. (1979) on the Oromo; De Heusch (1985) on the Zulu; Spencer (2014) on the Maasai; Foner and Kertzer (1978) on the Kikuyu; McCluskey (1984) on the Iteso; Dyson-Hudson (1963) on the Karimojong; Bohannan (1964) on the Tiv. Population estimates were taken from the most recent census data for each country.

Panel A. Uganda



Panel B. Kenya

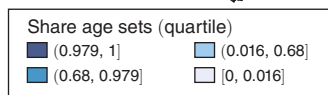
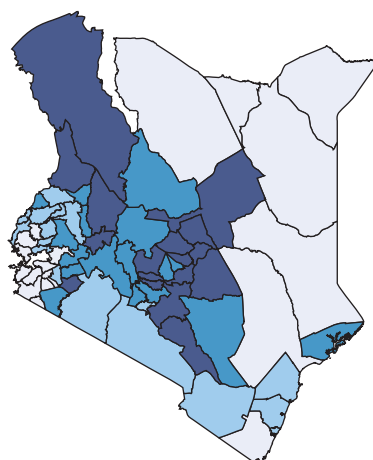


FIGURE 1.

Notes: This map displays district boundaries in Uganda (panel A) and county boundaries in Kenya (panel B), as well as country borders in bold. Regions are color-coded based on the share of Demographic and Health Survey (DHS) households in each district that are members of age set societies. Districts are divided into four quartiles based on the share of the population comprised of members of age set societies, with darker colors corresponding to higher shares. The data are from the most recent geo-coded wave of the DHS in each country; in Kenya this is the 2014 round and in Uganda it is the 2016 round. The underlying shapefiles are from data.world (2018) and Minnesota Population Center (2024) respectively.

B. Our Sample: Age Sets in Kenya and Uganda

Our empirical analysis focuses on policies implemented in Kenya and Uganda. For the societies in these regions, we determined whether or not age-based organization is the dominant form of social organization using the definition from Radcliffe-Brown (1929).¹² Data on the presence of age set organization is not available from standard datasets, including Murdock's (1967) *Ethnographic Atlas* (used frequently in empirical studies of social structure, e.g., La Ferrara 2007; Enke 2019; Ashraf et al. 2020; Bau 2021); thus, we determined ourselves whether or not each ethnic group is an "age set society" on the basis of extensive secondary source analysis. In online Appendix A.1, we list all ethnic groups in our sample, whether or not each is an age-based society, and the set of sources used to make that determination. We do not include an ethnic group in our sample unless we were able to determine with certainty whether it is an age-based society or not. In our analysis of Kenya's HSNP (Section II), we were able to code our age set variable for 99 percent

¹²It is possible for an ethnic group to have both age sets and a kinship structure. While many groups have either a strong lineage structure *or* a strong age-based structure, others exist along a spectrum. In online Appendix A.1 we describe in detail and provide clear examples of our coding strategy for determining which of the two forms of organization is dominant.

of the sample, and in our analysis of Uganda's pension program using DHS data (Section III), we were able to code our age set variable for 85 percent of the sample.

Figure 1 documents the geographic variation of age set organization in Kenya and Uganda. The figures are constructed using individual-level data from the latest round of the Demographic and Health Survey (DHS) in each country that contains both district identifiers and information about each respondent's ethnicity (Uganda Bureau of Statistics 2016 for Uganda and Kenya National Bureau of Statistics 2014 for Kenya). Each district is colored based on the share of surveyed households that we identify as belonging to an age set society. Members of age set societies make up 51 percent of the households in Kenya and 31 percent of the households in Uganda. While here we show district-level variation, in our empirical analysis, we use only within-district variation in ethnicity and age set organization; we display the aggregated data here for visualization. The sources that we use to determine whether each ethnic group has age sets are historical, and as a result, Figure 1 should be interpreted as capturing the share of individuals who are members of ethnic groups in which age sets were a traditionally important social unit. Nevertheless, we also use the DHS to investigate the extent to which one age set custom has persisted today. A key feature of age set societies is an initiation ritual (Radcliffe-Brown 1929, p. 21); in east Africa, the initiation ceremony often involves circumcision and takes place at around the age of puberty (Handley 2009; Spencer, 2014). Data on circumcision rituals are collected systematically by the DHS and allow us to directly investigate the prevalence of initiation rituals across ethnic groups. If age set customs are practiced today, we would expect that in age set societies: (i) the timing of circumcision is coordinated and (ii) circumcision is more likely to take place around the time of puberty.

Figure 2 displays a histogram of the age of circumcision for all DHS respondents, separately in societies with and without age sets; the mean and standard deviation of each distribution is reported at the bottom of the graph. Men in age-based societies are less likely to be circumcised at birth, more likely to be circumcised during the age of puberty, and the standard deviation of distribution of the age of circumcision is lower, suggesting coordination. These features of the data exist both in rural and urban settings when we plot the distributions separately (see online Appendix Figure A1). We interpret this pattern as a reassuring check that, consistent with qualitative accounts, age set ceremonies are prevalent today and discernible in our data.

C. The Function and Origin of Age Sets

The determinants of age set organization have not been a major emphasis of anthropological or ethnographic work, which has instead focused on describing and understanding the functioning and consequences of age sets. However, while there is no broadly accepted theory on the origins of age set organization, several hypotheses have been proposed.

In several contexts, age sets are used to organize men in battle; this has led LeVine and Sangree (1962) to hypothesize that age set organization began as a military innovation in response to the threat of external conflict. Consistent with this hypothesis, Zulu age set organization increased in importance during the rule of Shaka Zulu, a famous leader and military strategist who organized warriors into barracks

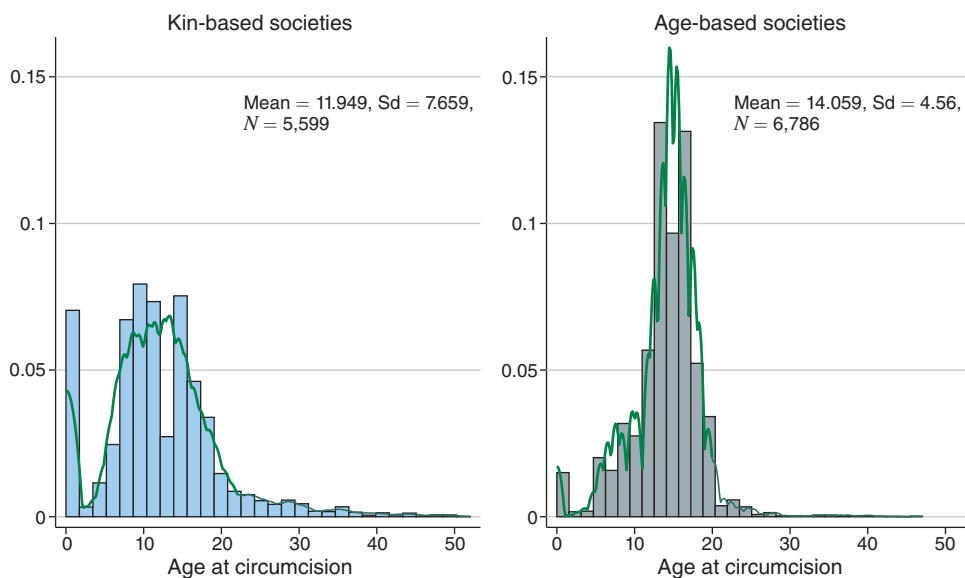


FIGURE 2.

Note: Histograms display the age of circumcision for all male respondents in the 2014 and 2016 DHS surveys for Kenya and Uganda respectively, separately for households from kin-based societies (left) and households from age set societies (right).

and fighting regiments based on their age set (Eldredge 2014). Among several east African age-based societies, particularly herding societies, the role of age sets in fighting remains particularly important; Morton (1979, p. 90) notes, “[T]he age set system represents the principal means for organizing manpower for the purpose of protecting the group against forcible intrusions from the outside and conducting offensive warfare against neighboring groups.”

Another possibility is that age set organization developed as a strategy to regulate political succession. In some age set societies, each cohort has a unique political role as part of an age-based hierarchy; the age set system determines who (i.e., which cohort) has political and decision making authority (Dyson-Hudson 1963). Third, the age set system is often related to marriage institutions and gender roles. An individual’s age set, and the marriage status of members of older adjacent age sets, can determine whether or not he is permitted to marry and, at times, his right to have children (Morton 1979). Recent ethnographic work from Kenya suggests that the age set system remains a male-dominated institution, and that the social network opportunities it provides are not open to women (Handley 2009).

A final view in the anthropological literature suggests that there are no clear causal factors leading to the development of age set organization; instead, age sets emerged through a complex evolutionary diffusion process likely originating in the region of the Oromo in Ethiopia (Beckingham and Huntingford 1954; Hinev 2012). This hypothesis follows a growing set of work arguing that social structure in general emerges in ways that can be idiosyncratic and not systematically related to any set of causal factors (Smith 1956; Salzman 1978; Kelly 1983). This

hypothesis is consistent with Moscona et al. (2020), who find no clear social, political, or geographic determinants of segmentary lineage organization. The idea that age set institutions could have diffused historically across space is one advantage to our within-country focus, comparing the behavior of members of societies with and without age sets in the same small geographic area.

We investigate the plausibility of each of these hypotheses in our sample of ethnic groups from Kenya and Uganda. We compile a range of ethnicity-level characteristics, from Murdock's (1967) *Ethnographic Atlas* and several other sources described in online Appendix A.2. We do not find evidence of systematic differences between age set and kin-based societies across a broad range of characteristics. Our 29 ethnic groups appear similar across our measures of violent conflict and herding, the transfer of political power, characteristics of marriage and inheritance, the role of women, and precolonial development.¹³ These estimates are described and reported in online Appendix A.3.

D. Age Sets and Economic Ties: Qualitative Evidence

Numerous ethnographic studies have documented that age set organization shapes patterns of economic interaction. Qualitative accounts suggest that members of age set societies have both stronger economic ties to individuals in their age cohort and weaker ties to members of other generations, including within their own family.

In one early study, Eisenstadt (1954, p. 101) notes, "The relations between age mates [involve] general and permanent obligations of co-operation, solidarity, [and] mutual help, closely resembling, in this respect, the family and kinship." Age sets play a central role in determining individuals' economic relationships and sense of identity. In the words of Dyson-Hudson (1963, p. 376), whose work focuses on the Karimojong cluster of ethnicities in Uganda, which form part of our sample, "An age set has members [...] who are equally uniformly identified by a single shared name, community of status, and a collective role." The fact that members of an age set share a singular identity is common to age set societies throughout sub-Saharan Africa. Morton (1979, p. 82) avers that the "highest correspondence [across east African age sets societies] exists in the manner in which egalitarian behavior among age set coevals is encouraged."

One of the most well-studied age set societies is the Maasai, who live between Kenya and Tanzania. Spencer (2014) describes that redistributing resources with age set members is both widespread and expected. During the initiation period, members of Maasai age sets "cultivate and parade an ethos of sharing," which is characterized by "an excessive display of 'group indulgence,' opposed to any suggestion of self-interest" (Spencer 2014, p. 45). Sharing of resources within the age set is not limited to *young* men, "[Later in life, Maasai men] share milk, meat, and the marital bed with age mates" (Spencer 2014, p. 46). Age set members also often

¹³We do find a statistically significant difference between societies with and without age sets in the case of jurisdictional hierarchy. It is possible that this relationship is due to random chance, given the number of separate hypotheses that we check (see online Appendix Table A4). Nevertheless, we document that all of our results are very similar after accounting for differences in levels of jurisdictional hierarchy at the ethnicity-level.

host each other for days at a time, and Maasai households have specific rooms that are meant for visiting age mates (Hodgson 1999).

Age sets take on a similar role among several ethnic groups in northern Kenya, the setting of the first part of our empirical analysis. Contemporary field notes by Handley (2009) document that among these age set societies (including the Samburu, Rendille, and Turkana) within-cohort ties are economically important. She paraphrases one interviewee:

[T]hose who are rich and those who are poor [in the age set] come together so that they may share milk, meat, tea, sugar, etc. so it puts everyone as equal. [I]t is then that you can't differentiate between rich and poor because they are covering each other. Bonding within the age set helps because the morans [age mates] can rely on each other even if you have no money or social problems.

Thus, in age set societies, within-cohort economic relationships are strengthened as individuals rely on members of their age cohort for support.

At the same time, according to ethnographic accounts, individuals in age set societies tend to rely *less* on family and intergenerational relationships. Maasai men often do not consider individuals outside their immediate family as part of their "lineage," as "it is toward age mates that men look for support rather than brothers or close patrilineal kin" (Spencer 2014, p. 48). The lack of intergenerational support is noted explicitly as one of the key functions of northern Kenyan age sets, which exist in part "so that the pressure is taken off the mother's house" (Handley 2009). According to one interviewee from the Samburu, "The group of morans [age mates] are the ones who have more responsibilities to each other, and *they are bound more to their fellow moran than they are to their brothers or sisters*" (Handley 2009, emphasis not in original). Occasionally, the relationship across cohorts and generations is not only weak, it is adversarial; Foner and Kertzer (1978) document that across age set societies, conflict sometimes breaks out between age sets during periods of initiations since age sets in a position of power are reluctant to relinquish their status.

A range of studies also suggests that the presence of age sets is associated not only with weaker relationships between members of the same extended family but also weaker relationships between members of the same immediate family. Bohannan (1964) writes that among the Tiv, the social bond and sense of obligation among age mates can be stronger than those even between parents and children. Legesse (1979, p. 63) notes that the "very common pattern of age set alternation that has been recorded by many ethnographers has the effect of dissociating fathers and sons" (see also Gulliver 1953; Turner 1955). This includes certain forms of "ritual estrangement" between parents and children during age set initiation, which takes place around the time of puberty. Moreover, Legesse (1973, p. 113) finds that among the Borana, mothers "begin to treat [their sons] in the same manner that they would treat their in-laws. [...] [W]hat started out as an intimate mother-son bond turns into formal estrangement [...] that is deliberately cultivated over the years." Particularly relevant for our study of the spillover effects of pension programs, Spencer (2014, p. 63) writes describes the relationship between Maasai elders and their younger family members, noting that in

the view of younger family members “elders are self-indulgent and mean hypocrites, who put their own selfish interests above those of their families and have a terrible curse over younger people.” Thus, in age set societies, ties between family members are anecdotally weaker, even ties between generations in the same immediate family.¹⁴

Together, these accounts suggest that age set societies have very different structure of allegiance and support than those typically analyzed. The dominant social group is the age cohort, and these horizontal ties are more important than kin-based relationships. The next sections document empirically how this distinct form of social organization affects economic interactions and the distributional effects of development policy.

II. Social Structure and Financial Ties: Experimental Evidence

This section analyzes a randomized evaluation of a large-scale cash transfer program—the Hunger Safety Net Program (HSNP)—in northern Kenya. The experimental design allows us to estimate both within-cohort and across-generation spillover effects of the unconditional cash transfer, in both age set and kin-based societies. We document strong within-cohort spillover effects in age set societies that are absent in kin-based societies. We also find within-clan and intergenerational spillover effects in kin-based societies that are absent in age set societies. These results indicate that local social organization shapes modern financial ties and the distributional impact of development policy.

A. Setting

Kenya’s Hunger Safety Net Program (HSNP) is a large-scale cash transfer program in northern Kenya.¹⁵ In its first phase, which took place between 2009–2012, it reached over 69,000 households; after 2012, it expanded to include 374,000 households, corresponding to over 2.1 million individuals. The program’s aim is to reduce extreme poverty by means of regular and substantial cash transfers. While the program is now widespread in the country, it was first piloted in 2009 in the regions of Mandera, Marsabit, Turkana, and Wajir, displayed in Figure 3.

The government of Kenya, together with Oxford Policy Management, selected 48 sublocations to conduct a randomized policy evaluation (see Merttens et al. 2013).¹⁶ The 48 sublocations were randomly assigned to either the treatment or control group. Households in both treated and control sublocations were interviewed at baseline (2009) and follow-up (2011); eligibility to receive the cash transfer was determined based on one of three targeting mechanisms, randomized across both treatment and

¹⁴ Additional ethnographic work describes how within-household arrangements also differ between age and kin-based societies. For example, Hodgson (1999) describes how members of Maasai households have rooms where visiting age-mates of the male head of household could stay. Women sleep in an inner room, while men would stay in an outer room, often with a visiting age mate. Gulliver (1958) describes how Turkana men live near several members of their age set and often participate in activities, including meals, with age set members rather than immediate family members.

¹⁵ The raw data and documentation are available at: <https://microdata.worldbank.org/index.php/catalog/1915>.

¹⁶ A sublocation is a super-set of the village.

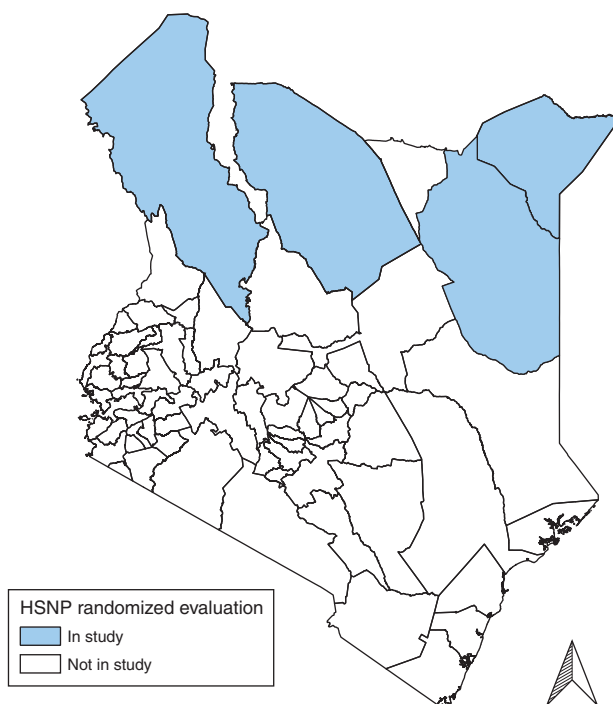


FIGURE 3.

Notes: Map showing the counties in Kenya included in the Hunger Safety Net Program (HSNP) randomized evaluation. These districts are (from left): Turkana, Marsabit, Wajir, and Mandera. The shape file used to construct the map was downloaded from Minnesota Population Center (2024).

control groups.¹⁷ Eligible households were identified in both treatment and control groups, but only eligible households in the treatment group began receiving transfers following the baseline survey; transfers were withheld from eligible households in the control group until after the first follow-up survey in 2011. The cash transfer consisted of KSh 2,500 (~US\$23) distributed every two months, corresponding to roughly 25 percent of annual in-sample household income. Substantial care was taken to ensure that only the intended recipients received the cash transfer; in an end line survey, less than 2 percent of intended beneficiaries reported issues with receiving their payment, and biometric fingerprinting was used to make sure that the funds could not be taken by nonbeneficiaries.¹⁸

¹⁷The targeting mechanisms were community based targeting, in which the community selected households to get the transfers; dependency ratio targeting, in which households were selected based on characteristics of household health and nutrition; and a social pension program, in which households were selected on the basis of the number of household members over the age of 55. We include all targeting mechanisms in our main analysis, and document that all findings are robust to the inclusion of targeting code by ethnicity fixed effects.

¹⁸The payment system is described in detail in Oxford Policy Management (2011a); for example, “Payments are made using a Smartcard. The Smartcards contain a microchip with the biometric fingerprints of the two nominated recipients of each beneficiary. [...] Cash is transferred electronically to the Smartcard and can be redeemed at any time at any of the agents using fingerprint scanning on Point of Sale devices (POS). Where electricity is not available, the POS are run by solar power” (p. 13).

A crucial feature of this policy experiment is that there are members of age-based and kin-based societies in both eligible and noneligible subgroups and in both treatment and control sublocations. We are unaware of any other randomized evaluation with members of both kin-based and age-based groups in the participant population, as well as all necessary data for the estimation of policy spillover effects (see Section B). This heterogeneity in participant ethnicity makes it possible to estimate spillover effects from the cash transfer program separately for members of societies with and without age sets.

B. Data

This section describes the data collected during the HSNP evaluation that we use in our empirical analysis. First, using each home language listed in the HSNP baseline survey (Oxford Policy Management 2010), we determine whether or not age set organization is the dominant social form for all households in the sample. This was done by hand from a range of secondary source material about the social structure of ethnic groups in northern Kenya, described in online Appendix A.1. The languages reported in the survey have a one-to-one correspondence with ethnicity, which allows us to straightforwardly link the two.

Second, the baseline HSNP dataset reports treatment status of each sublocation and eligibility status of each household, as well as the village and sublocation in which each household is located (Oxford Policy Management 2010). This allows us to identify the treatment status of each household. The baseline data also report the age of each household member, which is necessary for our estimation of cohort-level spillover effects.

Third, the baseline (Oxford Policy Management 2010) and follow-up surveys (Oxford Policy Management 2011b, 2012) report a range of household and individual-level information that allow us to construct the main dependent variables. Household-level consumption information is reported, which we use to identify consumption spillovers. Expenditure is broken down by purpose; in our empirical analysis, we rely primarily on estimates of total spending, spending on food, and spending on education. The HSNP also reports information on the weight and height of all children under the age of five, which we use to estimate spillover effects on young children.

Finally, the baseline survey includes a range of additional information that we use to construct control variables to investigate the robustness of our findings (Oxford Policy Management 2010). These covariates include information about health status, religion, marriage status, educational attainment, wealth, and a range of individual-level demographic information on each household member.

C. Within-Cohort Ties

This section investigates spillover effects within age cohorts. Our main hypothesis is that in age set societies, there are large spillover effects of the random cash transfers within the age cohort and that these effects are smaller or absent in kin-based societies.

Defining Age Cohorts.—We first construct a proxy for the share of people in an individual i 's “age set” that was eligible for the cash transfer program, Share Cohort Eligible (“SCE”):

$$(1) \quad SCE_{ihaev} = \left(\sum_{\substack{i' \neq i \\ i' \in a \times e \times v \times f(i)}} Eligible_{i'} \right) / \left(\sum_{\substack{i' \neq i \\ i' \in a \times e \times v \times f(i)}} Eligible_{i'} + NotEligible_{i'} \right)$$

where i indexes individuals, a indexes age cohorts, e indexes ethnic groups, v indexes villages and $f(i)$ denotes the gender of i . In words, SCE_{ihaev} is the share of eligible people in an individual's age cohort—defined as being the same age and gender, residing in the same village, and being from the same ethnic group—excluding the individual himself.¹⁹ We include only individuals of the same gender as the respondent since in this context, age sets involve predominately males and when female age organization exists, it is distinct from the male system (Handley 2009). Finally, when we focus on household-level consumption we assign each household the age set treatment of the main provider for the household. That is, we let $SCE_{haev} = SCE_{ihaev}$ where i is the main provider of household h .²⁰

Empirical Model.—We estimate the effect of changes in cohort-level income on consumption in age set versus kin-based societies. To do this, we exploit the fact that the experiment generated variation across cohorts in total transfer eligibility, but transfers were only delivered to those cohorts in sublocations randomly assigned to the treatment group.

The empirical strategy and main result can be described graphically by the bars reported in Figure 4. We present the correlation between SCE_{haev} (i.e., cohort-level eligibility) and (log of) consumption separately for four groups within the noneligible (i.e., spillover) population: (i) age set societies in control locations, (ii) age set societies in treatment locations, (iii) kin-based societies in control locations, (iv) kin-based societies in treatment locations. If our empirical design is valid, we expect no significant correlation between cohort-level transfer eligibility and consumption in control sublocations, where cash transfers were never delivered; this is precisely what we find in columns 1 and 3. If our main hypothesis is true, in treatment locations we expect cohort-level eligibility to affect consumption spending in age set societies (ii) but not in kin-based societies (iv). This is precisely the pattern we find in the data, a first indication of large within-cohort spillovers in age sets societies that are absent in kin-based societies.

¹⁹ While for some groups in our sample, like the Turkana, we know that age sets are initiated each year, in others an age set can contain individuals who are not exactly the same age. Unfortunately, we do not have data on the exact timing of initiation ceremonies; however, in all age set societies, individuals in the same cohort are part of the same age set. Thus, we use the above single-year measure as a conservative proxy for an individual's age set throughout the empirical analysis. Consistent with the fact that in some groups age sets include more than one birth year, Figure 5 shows some evidence of spillover effects to nearby cohorts.

²⁰ The results are all very similar and reported in the online Appendix Table A5 if we instead use age of the household head to assign each household to a cohort.

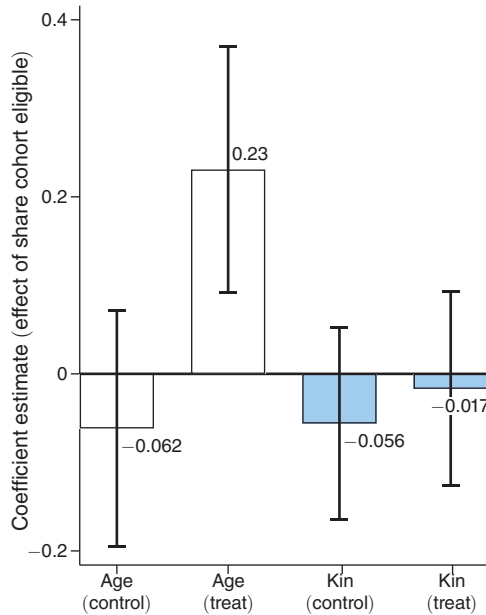


FIGURE 4. COHORT-LEVEL ELIGIBILITY AND CONSUMPTION, SEPARATELY BY TREATMENT STATUS AND SOCIAL STRUCTURE

Notes: Each bar reports the coefficient estimate of the relationship between a main provider's household consumption spending and SCE_{haev} . The regression is estimated on the subsample noted at the bottom. From left, these are: age set societies in control sublocations, age set societies in treatment sublocations, non-age set societies in control sublocations, non-age set societies in treatment sublocations. Interviewer and targeting code by ethnicity fixed effects are included in all specifications. 95 percent confidence intervals are reported.

Next, we turn to our more general econometric model. To estimate within-cohort spillover effects of cash transfers, we rely on the following triple-difference estimating equation:

$$(2) \quad y_{haev} = \gamma_1 \cdot (SCE_{haev} \cdot \mathbf{1}_{v(c)}^{Treat} \cdot \mathbf{1}_e^{AgeSet}) + \gamma_2 \cdot (SCE_{haev} \cdot \mathbf{1}_{v(c)}^{Treat} \cdot \mathbf{1}_e^{Kin}) \\ + \gamma_3 \cdot (SCE_{haev} \cdot \mathbf{1}_e^{AgeSet}) + \gamma_4 \cdot (SCE_{haev} \cdot \mathbf{1}_e^{Kin}) + X'_h \Gamma + \varepsilon_{haev},$$

where h indexes households, a age cohorts, e ethnic groups, and v villages, and c sublocations. The dependent variable is (log of) expenditure of the household of main provider i , and SCE_{haev} is defined as in equation (1), constructed for the main provider of household h ; $\mathbf{1}_{v(c)}^{Treat}$ is an indicator that equals one if sublocation c was assigned to the treatment group; $\mathbf{1}_e^{AgeSet}$ is an indicator that equals one if e is an age set society and $\mathbf{1}_e^{Kin}$ is an indicator that is equal to one if ethnicity e is a kin-based society. The model is also fully saturated with all components of the triple interaction (excluded from equation (2) for simplicity).

The coefficients of interest are γ_1 and γ_2 . These capture the effect of the cohort-level income shock on consumption spending, separately for members of age set and kin-based societies. The main hypotheses are that $\gamma_1 > 0$ and furthermore

TABLE 1—AGE COHORT SPILLOVER EFFECTS ON CONSUMPTION SPENDING

	log total consumption spending				Total cons. spending
	(1)	(2)	(3)	(4)	(5)
<i>Panel A. Full sample</i>					
<i>Share cohort eligible</i> $\times \mathbf{1}^{Treat} \times \mathbf{1}^{Age Set}$	0.258 (0.107)	0.339 (0.101)	0.279 (0.0976)	0.287 (0.105)	2,481 (966.9)
<i>Share cohort eligible</i> $\times \mathbf{1}^{Treat} \times \mathbf{1}^{Kin}$	−0.00613 (0.0761)	0.00364 (0.0798)	0.0218 (0.0799)	0.0315 (0.0799)	−815.3 (619.9)
<i>p</i> -value, $\gamma_1 = \gamma_2$	0.0550	0.0116	0.0528	0.0619	0.0052
Mean at baseline	7.32	7.32	7.32	7.32	8,026
R^2	0.471	0.628	0.682	0.684	0.673
Observations	713	646	646	643	643
<i>Panel B. Males only</i>					
<i>Share cohort eligible</i> $\times \mathbf{1}^{Treat} \times \mathbf{1}^{Age Set}$	0.365 (0.116)	0.443 (0.103)	0.344 (0.0965)	0.377 (0.105)	4,076 (1,036)
<i>Share cohort eligible</i> $\times \mathbf{1}^{Treat} \times \mathbf{1}^{Kin}$	−0.00234 (0.0723)	0.0194 (0.0831)	0.0487 (0.0834)	0.0499 (0.0874)	−698.9 (700.4)
<i>p</i> -value, $\gamma_1 = \gamma_2$	0.0111	0.00129	0.0218	0.0150	0.0003
Mean at baseline	7.35	7.34	7.34	7.34	8,272
R^2	0.487	0.663	0.703	0.710	0.689
Observations	603	548	548	545	545
Interviewer FE	Yes	Yes	Yes	Yes	Yes
Targeting code $\times$ ethnicity FE	Yes	Yes	Yes	Yes	Yes
Age $\times$ ethnicity FE	No	Yes	Yes	Yes	Yes
Additional controls	No	No	Yes	Yes	Yes
Age set $\times$ cohort eligibility FE	No	No	No	Yes	Yes

Notes: The unit of observation is a household. In panel A, we include the full sample of nonbeneficiary households and in panel B we include only households with a male main provider. The dependent variable is log of per capita monthly consumption spending in columns 1–4 and raw per-capita monthly consumption spending in column 5. Additional controls include gender, age, indicators for marriage, employment, disability, educational attainment, and religion, the number of household members, and a poverty index. Standard errors are clustered at the sublocation level.

that $\gamma_1 > \gamma_2$. That is, in age set societies, resources spill over within age cohorts and do so more than in societies without age sets.²¹ We also explore the robustness of our baseline findings to the inclusion of additional fixed effects and control variables, embodied in X'_h .

Results.—Table 1 reports our baseline estimates of equation (2). In panel A, the sample includes all main providers in the sample, and in panel B it is restricted to male main providers; in northern Kenya, age sets are a predominately male institution and we expect the results to be driven by this subsample (Handley 2009).²²

²¹Of additional interest are γ_3 and γ_4 , which capture differential selection of particular cohorts into HSNP eligibility in societies with and without age sets. If consumption is correlated within age cohorts, we would expect these terms to be negative since individuals with low levels of consumption are more likely to be in cohorts with a large number of transfer-eligible members. If consumption is *more correlated within age cohorts in age set societies compared to kin-based societies*, we would expect $\gamma_3 < \gamma_4$. We find evidence that both hypotheses are true (see online Appendix Figure A3). These findings motivate our triple difference design that directly accounts for these selection effects and makes it possible to compare cohorts with the same eligibility status across sublocations that were randomly assigned to the treatment or control group.

²²Moreover, female main providers are a very selected sample. Fifty-seven percent of female main providers are widowed or divorced, compared to 4 percent of male main providers.

In columns 1–4, the dependent variable is log of per-capita monthly consumption spending, and in column 5 it is total per-capita consumption spending.

In the first column, we control only for interviewer and targeting code by ethnic group fixed effects, as well as the baseline value of the dependent variable. These controls fully absorb any average differences across ethnic groups, separately for each arm of the transfer program. We find strong evidence of within-cohort spillover effects in age set societies ($\gamma_1 > 0$) and no evidence of within-cohort spillover effects in kin-based societies ($\gamma_2 = 0$). γ_1 is larger in magnitude focusing on the male subsample (panel B). The remaining columns of Table 1 show that the findings are very similar after including additional fixed effects and controls. In column 2 we control for age by ethnicity fixed effects, to fully absorb any differences in life cycle consumption patterns across ethnic groups. In column 3, we add a broad range of individual and household-level controls, listed in the table notes, and in column 4 we control for the age set indicator interacted with fixed effects in the cohort level shock, in order to flexibly capture any level differences in cohort eligibility across ethnic groups. The estimates are very similar in both cases. Finally, in column 5 we document that the results are similar if we use total consumption instead of log-transformed consumption as the dependent variable. Consistent with our hypothesis, γ_1 is substantially larger than γ_2 in all specifications, and the p -value of the difference between γ_1 and γ_2 is reported below each set of coefficients.

Our estimates suggest that a one standard deviation increase in age cohort cash transfers leads to a roughly 0.3 standard deviation increase in consumption spending in societies with age sets, and zero effect in societies without age sets. Stated differently, for an individual in the noneligible (i.e., spillover) group in an age set society, every additional cohort member treated by the cash transfer program leads on average to a US\$16 increase in yearly consumption spending and a US\$23 increase in yearly consumption spending when we restrict attention to the all male sample (panel B). This corresponds to roughly 15 percent of the size of the cash transfer, suggesting that a large share of the transfer is reallocated within the cohort; moreover, it corresponds to roughly 8 percent of the median in-sample income at baseline, suggesting that the within-cohort spillover effects of the cash transfer program are large relative to local income.

The findings are very similar and, if anything, more precise when we use food spending rather than total spending as the dependent variable. These estimates are reported in Table 2, which follows the exact same structure as Table 1. Across specifications, the difference between γ_1 and γ_2 is significant at the 1 percent level. This finding is consistent with the fact that the main purpose of the HSNP is to improve food security, and suggests that consumption spillover effects are concentrated on essential items in a population close to subsistence.

All estimates are largest for young and old age cohorts, with a more limited effect on individuals aged 30–50 (see online Appendix Figure A5). This pattern suggests that the age set structure continues to affect the consumption patterns of younger cohorts, where we observe large cohort-level spillovers, and will thus likely remain an important feature of society in the future. These heterogeneous effects across the life cycle are also consistent with the cohorts that have less income earning potential themselves (the young and old) relying disproportionately on the age set network.

TABLE 2—AGE COHORT SPILLOVER EFFECTS ON FOOD SPENDING

	log total consumption spending				Total cons. spending
	(1)	(2)	(3)	(4)	(5)
<i>Panel A. Full sample</i>					
<i>Share cohort eligible</i> $\times \mathbf{1}^{Treat} \times \mathbf{1}^{Age Set}$	0.261 (0.0992)	0.373 (0.0999)	0.314 (0.0981)	0.326 (0.103)	2,269 (833.8)
<i>Share cohort eligible</i> $\times \mathbf{1}^{Treat} \times \mathbf{1}^{Kin}$	−0.0218 (0.0891)	0.00949 (0.0869)	0.0209 (0.0776)	0.0390 (0.0847)	−592.4 (450.6)
<i>p</i> -value, $\gamma_1 = \gamma_2$	0.0428	0.00860	0.0255	0.0353	0.0036
Mean at baseline	7.04	7.04	7.04	7.04	6,039
R^2	0.352	0.550	0.619	0.624	0.591
Observations	713	646	646	643	643
<i>Panel B. Males only</i>					
<i>Share cohort eligible</i> $\times \mathbf{1}^{Treat} \times \mathbf{1}^{Age Set}$	0.348 (0.113)	0.502 (0.111)	0.385 (0.110)	0.422 (0.118)	3,635 (936.6)
<i>Share cohort eligible</i> $\times \mathbf{1}^{Treat} \times \mathbf{1}^{Kin}$	−0.0146 (0.0861)	0.0129 (0.0904)	0.0578 (0.0835)	0.0747 (0.0936)	−287.6 (610.9)
<i>p</i> -value, $\gamma_1 = \gamma_2$	0.0145	0.000822	0.0201	0.0182	0.00071
Mean at baseline	7.07	7.06	7.06	7.06	6,227
R^2	0.368	0.594	0.639	0.646	0.609
Observations	603	548	548	545	545
Interviewer FE	Yes	Yes	Yes	Yes	Yes
Targeting code $\times$ ethnicity FE	Yes	Yes	Yes	Yes	Yes
Age $\times$ ethnicity FE	No	Yes	Yes	Yes	Yes
Additional controls	No	No	Yes	Yes	Yes
Age set $\times$ share eligible in age cohort	No	No	No	Yes	Yes

Notes: The unit of observation is a household. In Panel A, we include the full sample of nonbeneficiary households and in panel B we include only households with a male main provider. The dependent variable is log of per capita monthly food spending in columns 1–4 and raw per-capita monthly food spending in column 5. Additional controls include gender, age, indicators for marriage, employment, disability, educational attainment, and religion, the number of household members, and a poverty index. Standard errors are clustered at the sublocation level.

Falsification Tests: We estimate a series of falsification tests to further build confidence in our interpretation of these findings and convey that they are driven by the age set system. For each individual, we compute the share of transfer-eligible individuals in the same village and (i) in cohorts 2–5 years away, (ii) in cohorts 6–10 years away, (iii) in cohorts 11–15 years away, and (iv) in cohorts 16–20 years away. We then estimate a versions of equation (2) replacing the share of transfer-eligible people in an individual's *own* cohort with the share of transfer-eligible people in the other cohort groups. In some cases, age sets include more than one cohort so we may expect consumption spillover effects from cohorts two to five years away; however, we hypothesize that they should be smaller than the spillover effects from an individual's own cohort and we do not expect to see any effects from cohorts greater than five years older or younger.

Our estimates of these spillover effects are reported in Figure 5. We report our estimate of γ_1 from each specification and show the results separately for total consumption (Figure 5, panel A) and food consumption (Figure 5, panel B). First, for comparison, we display the spillover effect from one's own cohort (leftmost bar). The remaining columns show the effect of transfer eligibility of individuals in the same

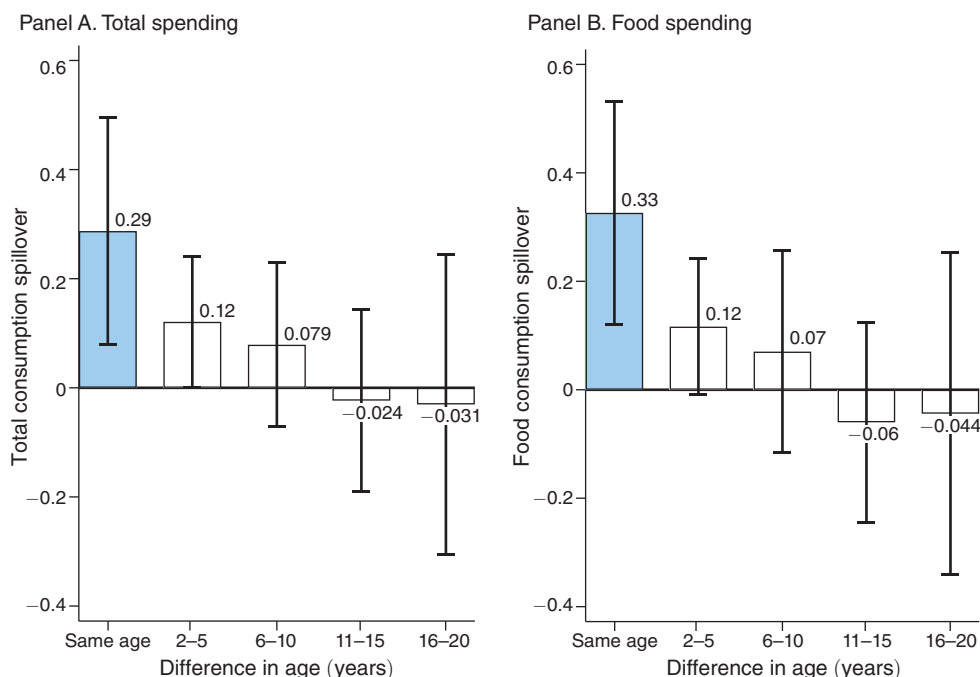


FIGURE 5. EFFECT OF OWN-COHORT VERSUS OTHER-COHORT SHOCKS

Notes: Estimates of equation (2) with the eligibility of a series of different cohorts used to compute the right hand side variables. The x -axis reports the cohort(s) relative to an individual's own cohort. The leftmost bar reports the spillover effect from an individual's own cohort, our baseline estimate. The dependent variable is (log of) total expenditure in panel A and (log of) food expenditure in panel B. All estimates are from the most conservative specification from Table 1 (column 4). Standard errors are clustered by sublocation and 95 percent confidence intervals are reported.

village but in other cohorts; while, as predicted, the effect is positive (albeit insignificant) for close cohorts, it is very close to zero for all remaining cohort groups. This pattern highlights that spillover effects in age set societies are driven by an individual's own cohort, and not greater income sharing overall within the village.

Sensitivity Analysis: Omitted Variables: We next investigate the robustness of the estimates to controlling flexibly for ethnicity-level characteristics other than age set organization; these estimates are reported in online Appendix Figure A4 and described in greater detail in online Appendix Section B.1. To compile ethnicity-level covariates, we match the ethnic groups present in our sample to the corresponding societies in Murdock's (1967) *Ethnographic Atlas*. We first flexibly control for each ethnicity's language group, thus homing in on within language-group variation in social structure. We then control directly for several measures of precolonial development. Our estimates remain similar, indicating that the results are driven by age set organization and not other omitted characteristics of the ethnic groups in our sample. Moreover, many additional ethnicity-level covariates are constant within our sample, which suggests that we are generally working with a comparable set of ethnicities that are similar across

several important dimensions.²³ Finally, a potential concern is that members of highly treated age sets spend more not because of actual transfers from members of their age set but because, anticipating transfers in the future, they begin to spend out of their savings. This does not seem to be driving the findings, however, because only 8 percent of the sample reports having any cash savings at baseline and the main findings are very similar if we exclude households with any savings from the sample (online Appendix Table A6).²⁴

Sensitivity Analysis: Inference: Our main estimates exploit differences in social structure across the full set of ethnic groups in the HSNP sample. This is a limited number of distinct groups, some of which comprise a relatively large share of the sample (e.g., the Somali make up 35 percent of the sample and the majority of the kin-based group). First, in order to make sure that the findings are not driven by any particular group, we replicate the baseline estimates after excluding each group from the sample one-by-one; in all cases, the coefficients of interest remain similar (see online Appendix Tables A7 and A8).

Second, to investigate this issue systematically, we conduct inference using resampling procedures. We generate placebo categorizations randomly assigning each ethnicity to either the age-based or kin-based group, both with and without replacement. The former (“bootstrapping”) provides the correct coverage under the assumption that uncertainty derives from our sampling a (relatively small) set of ethnic groups from a larger population of ethnic groups. The latter (“randomization inference”) provides the correct coverage under the hypothesis that uncertainty derives from the assignment of ethnic groups to the two types of social organization that we consider. We compute our estimates of $\gamma_1 - \gamma_2$ for each placebo assignment and estimate the randomization p -value as the share of placebo estimates larger than our baseline estimates. The distributions of placebo coefficients, as well as our actual estimates marked with a dotted line, are reported in online Appendix Figure A6. Using the bootstrap, when the dependent variable is (log of) total consumption expenditure, $p = 0.047$ using the full sample and $p = 0.043$ after restricting the sample to males. When the dependent variable is (log of) food expenditure, the p -values are $p = 0.046$ and $p = 0.042$, respectively. Randomizing without replacement, the four p -values are $p = 0.114$, $p = 0.048$, $p = 0.112$, and $p = 0.040$, respectively.

Finally, we show that the precision of the results is similar if we double-cluster standard errors by sublocation and by “solidarity group,” capturing the idea that errors may be correlated within the social group in which financial transfers takes place. We define the “solidarity group” as the age cohort in age-based societies and as the sub-clan in kin-based societies. Online Appendix Tables A9 and A10 reproduce our baseline results, with total spending and food spending as the dependent variables, respectively, using this alternative clustering strategy.

²³ Characteristics that are constant within our sample, according to Murdock’s (1967) *Ethnographic Atlas*, include: matrilocality/patrilocality, matrilineality/patrilineality, bride price customs, cousin marriage customs, and inheritance customs.

²⁴ One issue with this approach is that households may store savings as real assets and that these would not be captured in the survey question about cash savings. Nevertheless, the payment delivery mechanism makes it highly unlikely that individuals spend more because they falsely anticipate that they will receive a transfer in the future or because of disbursement to ineligible individuals. The delivery mechanism is described in Oxford Policy Management (2011a) as well as Section IIA, herein.

TABLE 3—AGE COHORT AND SUB-CLAN SPOILOVER EFFECTS ON CONSUMPTION SPENDING

	log total spending			log food spending		
	Full sample (1)	Full sample (2)	Males only (3)	Full sample (4)	Full sample (5)	Males only (6)
<i>Share sub-clan eligible</i> $\times \mathbf{1}^{Treat} \times \mathbf{1}^{Age Set}$	−0.127 (0.197)	−0.0359 (0.223)	−0.157 (0.247)	−0.0147 (0.258)	0.122 (0.291)	−0.0619 (0.313)
<i>Share sub-clan eligible</i> $\times \mathbf{1}^{Treat} \times \mathbf{1}^{Kin}$	0.248 (0.209)	0.363 (0.206)	0.404 (0.198)	0.378 (0.223)	0.480 (0.234)	0.503 (0.242)
<i>Share cohort eligible</i> $\times \mathbf{1}^{Treat} \times \mathbf{1}^{Age Set}$	0.387 (0.151)	0.507 (0.151)	0.542 (0.164)	0.312 (0.161)	0.424 (0.162)	0.461 (0.188)
<i>Share cohort eligible</i> $\times \mathbf{1}^{Treat} \times \mathbf{1}^{Kin}$	0.0106 (0.108)	0.00996 (0.100)	0.0272 (0.115)	−0.0253 (0.129)	−0.0232 (0.117)	−0.00143 (0.127)
<i>p</i> -value, <i>age cohort shocks</i>	0.043	0.008	0.011	0.102	0.030	0.045
<i>p</i> -value, <i>clan shocks</i>	0.202	0.191	0.067	0.254	0.337	0.148
Observations	436	430	371	436	430	371
R^2	0.686	0.716	0.716	0.644	0.670	0.666
Interviewer FE	Yes	Yes	Yes	Yes	Yes	Yes
Targeting code $\times$ ethnicity FE	Yes	Yes	Yes	Yes	Yes	Yes
Age $\times$ ethnicity FE	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	Yes	Yes	Yes	Yes	Yes	Yes
Clan FE	No	Yes	Yes	No	Yes	Yes

Notes: The unit of observation is a household. The dependent variable is log of per capita monthly consumption spending in columns 1–3 and log of food spending in columns 4–6. Additional controls include gender, age, indicators for marriage, employment, disability, educational attainment, and religion, the number of household members, and a poverty index. Standard errors are clustered at the sublocation level.

D. Kin-Based and Intergenerational Ties

The previous section documented large within-cohort spillover effects in age set societies, and found no evidence of within-cohort ties in the kin-based societies in our sample. While ethnographic accounts suggest that within-cohort links in age set societies substitute for intergenerational and family ties that prevail in kin-based societies, our results to this point are also consistent with members of age set societies sharing more income *in general*. In this section, we directly investigate the prevalence of within-lineage and intergenerational transfers across ethnic groups.

Since the HSNP does not report information about an individual's extended family, which would allow us to estimate spillover effects within the kin network, we use two alternative approaches to capture kin-based financial ties. First, we estimate consumption spillover effects within each *sub-clan*; sub-clan identity is self-reported by a subset of the sample and is a rough proxy for the extended family group. Second, we focus on the pension program targeting mechanism built into the randomized experiment and estimate the impact of random transfers to older generations on child nutrition. This analysis of intergenerational transfers motivates a more extensive analysis in Section III, where we study the rollout of Uganda's national pension program.

Table 3 reports estimates from an augmented version of equation (2) in which, in addition to the share of age cohort members eligible for the transfer interacted with treatment and kin- or age-based group indicators, we also include the share of sub-clan members eligible for the transfer interacted with treatment and kin- or age-based group indicators. While the results are less precise than our previous

estimates, likely due in part to the smaller sample size and imprecision of the sub-clan variable, there are three main conclusions.²⁵

First, as shown in Table 3, we find no evidence of spillover effects within the sub-clan in age-based societies (row 1); all coefficient estimates are insignificant and close to zero. Second, we find some evidence of positive spillover effects within the sub-clan in kin-based societies (row 2). While the significance level fluctuates across specifications, all estimates are all positive, large, and of comparable magnitude to the cohort-level spillover effects in age set societies. Moreover, the precision of the estimates is similar when we conduct inference using the resampling procedures described in Section IIC under "Results." The distributions of placebo estimates and randomization p -values are reported in online Appendix Figure A7; if anything, the randomization p -values for the difference between the effect of clan income shocks in age-based versus kin-based societies are slightly smaller using randomization inference. Finally, large within-cohort spillover effects remain apparent for age-based societies and not kin-based societies after accounting for sub-clan-level spillover effects from the transfer program (rows 3 and 4). These findings are a first indication that within-family transfers are present in the kin-based groups in our sample but absent in the age-based groups.

Next, we study intergenerational financial links in age set and kin-based societies, exploiting the pension program treatment arm in which transfers were given to all individuals in treatment locations over the age of 55. We restrict the sample to the set of eligible households in villages that were randomly assigned to the pension grant treatment arm of the experiment, and estimate the following equation:

$$(3) \quad y_{iev} = \xi_1 \cdot (\mathbf{1}_v^{Treat} \cdot \mathbf{1}_e^{AgeSet}) + \xi_2 \cdot (\mathbf{1}_v^{Treat} \cdot \mathbf{1}_e^{Kin}) + X'_i \Gamma + \varepsilon_{iev},$$

where i indexes either children in transfer-eligible households or transfer-eligible households as a whole (depending on the specification). Our main dependent variable is household education spending, which captures investment in younger generations.²⁶ ξ_1 captures the impact of the randomly assigned pension program on our proxies for investment in children in age set societies, and ξ_2 captures the same for kin-based societies. We hypothesize that $\xi_2 > 0$ and moreover that $\xi_2 > \xi_1$; if intergenerational ties are fully absent in age set societies, we would find that $\xi_1 = 0$.

Figure 6 reports estimates of ξ_1 and ξ_2 from equation (3); the unit of observation is the household and the dependent variables are (log of) education spending (panel A) and (log of) education spending as a share of total spending (panel B). In both cases,

²⁵ A potential concern is that reporting information about sub-clan affiliation in the baseline survey could be related to the economic importance of sub-clan organization. It seems likely that this would bias us away from finding any effect, if members of age set societies for whom sub-clans are least important are endogenously excluded from the sample. This could be the case if, for example, all members of kin-based societies report sub-clan membership, while only members of age set societies from whom sub-clans are most important report sub-clan membership. This reporting difference would bias our estimates away from our hypothesized result.

²⁶ In the online Appendix, we include the same analysis for the weight-for-height percentile of children under 60 months of age. There, we focus on weight-for-height rather than any anthropometric measures normalized by age because the age of children in months was not recorded; below, in our analysis of Uganda's pension program where we have more comprehensive measures of anthropometry, we document the robustness of our findings across a range of possible parameterizations.

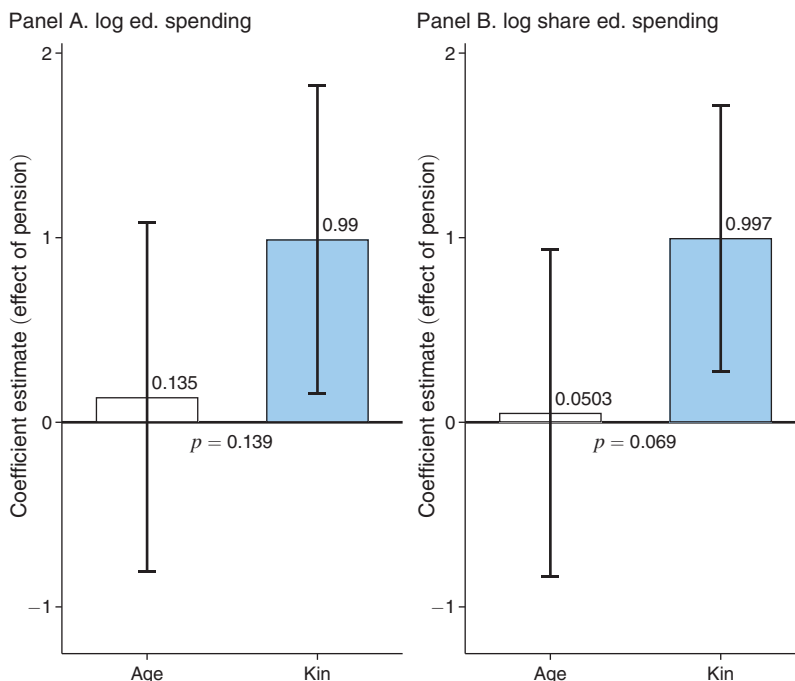


FIGURE 6. PENSION GRANTS AND INVESTMENT IN CHILDREN: AGE SET VERSUS KIN

Notes: Each panel reports estimates of equation (3); the left bar reports ξ_1 and the right bar ξ_2 . The dependent variable is listed at the bottom of each graph. All specifications include interviewer fixed effects, ethnicity fixed effects, age-by-age set fixed effects, and the full set of baseline controls. Standard errors are clustered by sublocation and 95 percent confidence intervals are reported.

we find no effect of the pension program in age set societies but a positive and significant effect in kin-based societies. Moreover, this difference between age-based and kin-based societies is entirely restricted to education spending (online Appendix Table A11). Forms of investment that benefit all generations do not respond differentially to the pension program.²⁷

Another strategy for measuring investment in children, which we return to in Section III, is to use data on child anthropometric outcomes, as a proxy for child nutrition. The HSNP survey collected anthropometric data at follow-up for a relatively small sample of children (roughly one-third of the baseline sample of children and of the sample of Figure 6); nevertheless, online Appendix Figure A8 investigates the relationship between the pension and child nutrition on this subsample. Here, the unit of observation is a child and the dependent variable is weight-for-height. While we find no difference between age-based and kin-based societies in the full sample of children (A8a), when restricting attention to male children we find that weight-for-height increases substantially in kin-based societies, but not age-based societies, during the pension program (A8b).

²⁷Online Appendix Table A11 also reports additional specifications documenting that the distinction between age and kin-based societies is even starker focusing on male main providers.

A remaining question is whether the results in Figure 6 are driven by (i) less of the pension money reaching the household budget in age set societies (i.e., because a larger share is transferred to individuals outside the household) or (ii) by the pension money being allocated differently within the household in age set societies. Online Appendix Figure A5 shows that in age set societies there is substantial redistribution across households with older main providers, suggesting some of the pension money exits the household. However, we find no differences in the effect of pension receipt on total household expenditure between age-based and kin-based societies, suggesting that the results in Figure 6 are not driven by average differences in interhousehold redistribution.²⁸ Instead, it seems likely that the elderly are able to spend the pension money on different household goods in the two groups, consistent with a broad range of work arguing that the identity of income recipients within the household can shape patterns of spending (see, for example Duflo 2003; Robinson 2012; Haushofer and Shapiro 2016). Building on this body of evidence, our results show that the relationship between the income recipient's identity and spending choices can vary widely across contexts and that this variation may be determined by differences in social organization.

III. Social Structure and National Policy: Uganda's Pension Program

The previous section exploited a randomized cash transfer experiment to document that age set societies have strong within-cohort economic ties, and comparatively weak kin-based ties. The latter finding was apparent in the context of the pension program treatment arm of the experiment, which led to substantial improvements in proxies for investment in children in kin-based societies and had no such effects in age set societies. Motivated by this finding, and the fact that the overall impacts of development policy can hinge on patterns of intergenerational ties and investment (Duflo 2003; Ashraf et al. 2020), this section investigates the intergenerational consequences of Uganda's national pension program and how its effects differ in age set and kin-based social structures.

A. Setting

In 2011, Uganda introduced the Senior Citizen Grant (SCG), a social pension program targeting all individuals aged 65 and older.²⁹ The pension program pilot consisted of US\$7.50 monthly transfers, which corresponds to around 20 percent of average monthly per-capita consumption. Rather than introduce the program nationally, the government of Uganda piloted the program in just 15 districts; the set of pilot districts are displayed in Figure 7. This partial rollout makes it possible to estimate the economic consequences of the program.

An explicit goal of the pension program was to improve the well-being of children, since the government of Uganda believed that pension recipients would invest

²⁸ When we estimate the difference in the marginal effect of pension eligibility on log of total household spending between members of age-based and kin-based societies, we find a coefficient of 0.19 (SE = 0.12), which is small in magnitude and statistically indistinguishable from zero.

²⁹ The age eligibility cutoff was 60 in the Karamoja region and we account for this throughout the analysis.

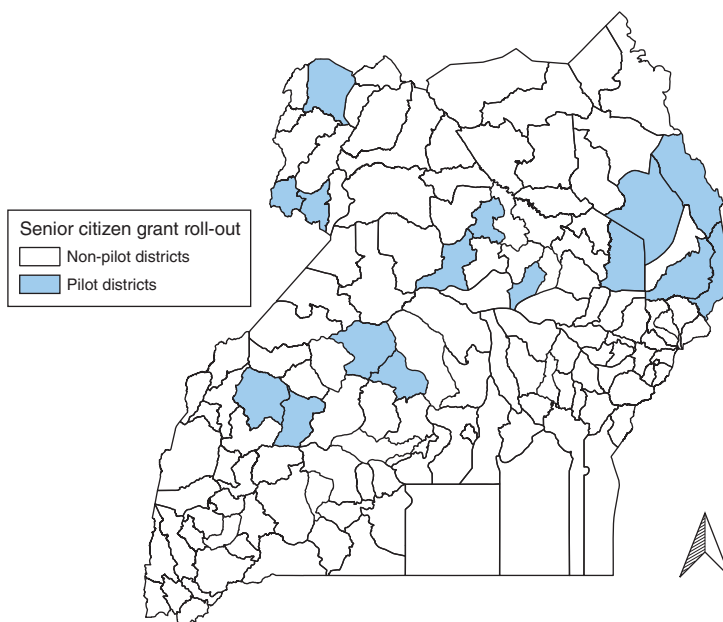


FIGURE 7.

Notes: This map shows Uganda district boundaries and identifies in blue the set of pilot districts in the 2011 pilot program of the Senior Citizen Grant (SCG). There are 110 districts included in our baseline sample after linking district and ethnicity-level information to the DHS. The shape file used to construct the map was downloaded from data.world (2018).

part of the transfer in grandchildren. The website of Uganda's Ministry of Gender, Labour and Social Development, which was in charge of implementing the pension program, reads:

Although the SCGs target older people, the grants benefit more than only the senior citizen beneficiaries. The SCGs have significant impact on development outcomes *as older people tend to invest a portion of their grant money in meeting their grandchildren's nutritional, health, education needs.*³⁰

The staggered introduction of the pension program, combined with the widespread prevalence of age set organization in Uganda, makes it an ideal setting to study how social structure shapes intergenerational economic ties and the spillover effect of major national policy.

B. Data

The main dataset for this analysis is Uganda's Demographic and Health Survey (DHS). Our primary analysis relies on the 2016 round of the survey (Uganda Bureau of Statistics 2016); we also use the 2006 round of the DHS, which was conducted

³⁰ See here: <https://socialprotection.go.ug/wp-content/uploads/2018/07/Disability-response-in-the-Senior-Citizens-Grant-1.pdf>.

prior to the introduction of the SCG, to conduct a placebo exercise (Uganda Bureau of Statistics 2006). For each household in the survey, the DHS reports information on household composition, including the age of all members, which we use to identify which households are exposed to the pension program. The survey also reports anthropometric data for all children under five years of age and schooling information for all school aged children, which are our proxies for investment in children.

All household members selected for the individual component of the survey report their ethnicity, which we match to the household data using unique household identifiers. Using self-reported ethnicity and a range of anthropological and ethnographic accounts, we determine whether age sets are present or not for as many ethnic groups as possible in the DHS sample; the details of this data collection process are discussed in online Appendix A.1.³¹ Members of age set societies comprise 29 percent of our sample.

Finally, we use the Ministry of Gender, Labour, and Social Development website to identify the districts that were included in the SCG pilot program (displayed in Figure 7).

C. Estimation Strategy

Measuring Potential Pension Exposure.—We construct a measure of “potential pension exposure” for all households in the sample, in both pilot and non-pilot districts. Our baseline potential pension exposure (*PPE*) measure is

$$(4) \quad PPE_h = \sum_{t=2011}^{2016} \left(\sum_{i \in h} \mathbf{1}_{iht}^{Age65+} \right),$$

where i indexes individuals, h indexes households, and t indexes years since 2011; $\mathbf{1}_{iht}^{Age65+}$ is an indicator that equals one if individual i in household h was at least 65 years old in year t ; we construct this indicator for all individuals in every household using household member age data from the DHS. Thus, this instrument for pension exposure captures the number of years of pension grants each household would have received if its household composition observed in 2016 were fixed during the preceding years.

Empirical Model.—In order to identify the effect of the pension program on children, we compare households with a given level of potential pension exposure across pilot and non-pilot districts. We conduct this comparison separately for households from age set societies and households from kin-based societies.

Mirroring our identification strategy for estimating spillovers in Kenya’s HSNP, our strategy and results can be illustrated by a simple graph. In Figure 8, we report the partial correlation between PPE_h and child weight-for-height across four

³¹ We were able to definitively determine whether or not 85 percent of the households in the sample belong to an age set society. For the remaining 15 percent, either ethnicity was not specified, or there was insufficient secondary source information to make this determination with certainty. Most un-coded ethnic groups are small and comprise less than 2 percent of the population.

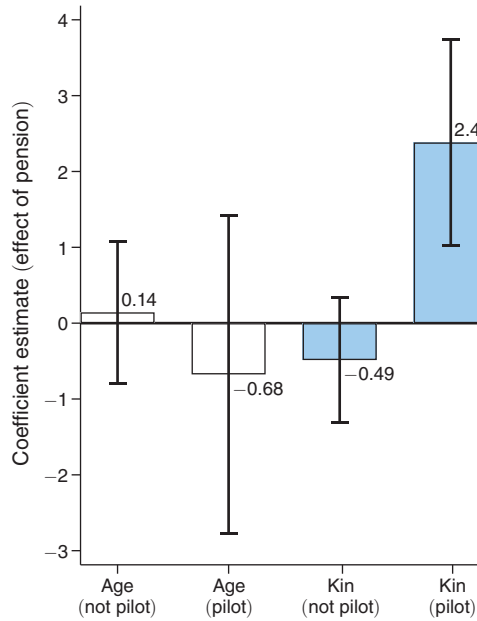


FIGURE 8. PENSION EXPOSURE AND CHILD NUTRITION, SEPARATELY BY PILOT STATUS AND SOCIAL STRUCTURE

Notes: Each bar reports the coefficient estimate of the relationship between a child's weight-for-height percentile and $PensionExposure_h$. The regression is estimated on the subsample noted at the bottom. From left, these are: age set societies in non-pilot districts, age set societies in pilot districts, non-age set societies in non-pilot districts, non-age set societies in pilot districts. District-by-ethnicity and interview month fixed effects are included in all specifications; 95 percent confidence intervals are reported.

subpopulations: (i) age set societies in non-pilot districts, (ii) age set societies in pilot districts, (iii) kin-based societies in non-pilot districts, (iv) kin-based societies in pilot districts. If our empirical design is valid, we expect no relationship between potential pension exposure and child nutrition in non-pilot districts where pension grants were not delivered; this is what we find in bars 1 and 3. Moreover, consistent with our main hypothesis, we find no effect of potential pension exposure on child nutrition among members of age set societies in pilot districts (bar 2), but a large positive effect of potential pension exposure on child nutrition among members of kin-based societies in pilot districts (bar 4).

Our more general econometric model is

$$(5) \quad y_{ihde} = \beta_1 \cdot (PPE_h \cdot \mathbf{1}_d^{Pilot} \cdot \mathbf{1}_e^{Kin}) + \beta_2 \cdot (PPE_h \cdot \mathbf{1}_d^{Pilot} \cdot \mathbf{1}_e^{AgeSet}) \\ + \beta_3 \cdot (PPE_h \cdot \mathbf{1}_d^{Pilot}) + \beta_4 \cdot (PPE_h \cdot \mathbf{1}_e^{Kin}) + \alpha_{ed} + \mathbf{X}'_i \Gamma + \epsilon_{ihde},$$

where h continues to index households and i indexes individuals. In the case of the main results, the sample is restricted to all children under 60 months of age (i.e., the sample for which anthropometric data were collected). d is the district in which the household is located and e is the household's ethnic group. PPE_h , potential pension exposure, is defined in equation (4). $\mathbf{1}_d^{Pilot}$ is an indicator that equals one

if a household is located in a pension pilot district, and $\mathbf{1}_e^{AgeSet}$ and $\mathbf{1}_e^{Kin}$ are indicators that equal one if a household is part of a kin-based or age-based ethnic group, respectively. Standard errors are clustered by district.

Our hypotheses are that $\beta_1 > 0$ and that $\beta_1 > \beta_2$. That is, we hypothesize that transfers to older household members benefit children in kin-based societies (where intergenerational ties are present) and does so *more* than in societies with age sets (where intergenerational ties are weak). Our estimate of interest is $\beta_2 - \beta_1$: the differential effect of the pension program on child nutrition in societies without versus with age sets. Further, if intergenerational ties in age set societies are completely absent, we would expect to find that $\beta_2 = 0$.

One potential shortcoming of this analysis is that we only observe a child's exposure to older family members if those older family members are in the same household.³² Thus, our baseline estimates do not capture differences in intergenerational relationships in general but rather differences in intergenerational relationships within the extended household. While most ethnographic work on the distinction between age-based and kin-based groups focus on interhousehold ties, a range of studies also describe differences in the relationships between members of the same immediate family or household (see Section ID). However, since estimates of equation 5 do not capture the effect of differences in interhousehold relationships, our empirical estimates from this section can be thought of as a more extreme test of our motivating hypothesis. In IID below, we also report estimates from the "double difference" analog to equation (5) that does not exploit variation in pension exposure across households and instead estimates differences in the average effect of the pension program on child nutrition in age-based versus kin-based societies. While exploiting coarser variation in pension exposure, these estimates combine the impact of within-household and cross-household intergenerational investments.

Balance Tests.—Our key identification assumption is that absent the pension program, differences between children in societies with and without age sets should be unrelated to pension exposure. To investigate the plausibility of this assumption, we estimate differences between households from societies with and without age sets in the non-pilot group. Results from this analysis are reported in online Appendix Table A12. First, we show that there is no correlation between age set organization and measures of potential pension exposure and household composition. These variables are all measured at the household level (panel A). Second, we show that there is no relationship between age set organization and our measures of child nutrition (panel B). Average child age is also balanced across both groups, which suggests that children from the two groups are not differentially selected into our sample due to different rates of infant mortality.

A remaining possibility is that, even though age set organization is not correlated with child nutrition in control districts, pension program exposure (i.e., the presence of older household members) may be differentially correlated with child nutrition in age set societies in the non-pilot group. It would be a concern,

³²The 2016 DHS survey from Uganda does not collect data on whether adults have living family members residing outside their household.

TABLE 4—EFFECTS OF UGANDA’S SCG ON CHILD NUTRITION IN SOCIETIES WITH AND WITHOUT AGE SETS

Dep. var.: weight for height percentile:	Full sample			Males	Females
	(1)	(2)	(3)	(4)	(5)
$Pension\ exposure \times \mathbf{1}^{Pilot} \times \mathbf{1}^{Kin}$	2.477 (0.605)	2.943 (1.008)	3.135 (1.098)	5.729 (1.428)	2.748 (1.195)
$Pension\ exposure \times \mathbf{1}^{Pilot} \times \mathbf{1}^{Age\ Set}$	−0.649 (0.965)	−0.818 (1.026)	−0.933 (0.951)	−2.440 (2.542)	0.107 (1.173)
Observations	4,112	4,107	4,107	2,020	1,976
R^2	0.130	0.131	0.203	0.240	0.241
District $\times$ ethnicity FE	Yes	Yes	Yes	Yes	Yes
Interview month FE	Yes	Yes	Yes	Yes	Yes
Age set $\times$ potential exposure FE	No	Yes	Yes	Yes	Yes
Age in months $\times$ gender FE	No	No	Yes	Yes	Yes
p -value, $\beta_1 = \beta_2$	0.006	0.009	0.007	0.008	0.122
Mean non-pilot	49.054	49.054	49.054	49.054	49.054

Notes: The unit of observation is the child. The sample includes all children in the 2016 round of Uganda’s DHS who are less than 60 months old. The dependent variable is the percentile of the weight-for-height distribution. Pension exposure (PPE) is constructed as in equation (6). $\mathbf{1}^{Pilot}$ is an indicator variable that equals one if the household is in a pilot district. $\mathbf{1}^{Age\ Set}$ is an indicator variable that takes value one if the household belongs to an age-based society. $\mathbf{1}^{Kin}$ is an indicator variable that takes value one if the household belongs to kin-based society. Standard errors are clustered at the district level.

for example, if in the control group, potential pension exposure is positively correlated with child health in kin-based societies but not age set societies. To test this directly, in panel C of online Appendix Table A12 we report the correlation between the interaction term between potential pension exposure and age set organization, and measures of child nutrition. The estimated coefficients are small and statistically insignificant in all specifications. Together, these findings suggest that it is unlikely that our results are driven by baseline differences between societies with and without age sets in either child nutrition or household composition, which could affect selection into pension exposure in ways that could bias the main estimates.

D. Main Results

Our main finding is that household exposure to the pension program improved child nutrition in societies without age sets, but had *no impact on child nutrition in societies with age sets*. We interpret these findings to suggest that intergenerational ties in age set societies are sufficiently weak that an income shock to the grandparent generation has no discernible impact on child well-being.

Our baseline estimates of equation (5) are reported in Table 4. Following Duflo (2003), we use child weight-for-height as our baseline anthropometric dependent variable, but show below that the findings are similar using alternative metrics. In column 1, we control only for ethnicity-by-district and interview month fixed effects. β_1 is positive and significant, suggesting that household pension exposure improved child nutrition in kin-based societies. However, β_2 is statistically indistinguishable from zero, and is significantly smaller than β_1 (p -value = 0.006). An additional year of pension receipt in a household without age sets, compared to one with age sets, increases child weight by roughly 0.15 standard deviations

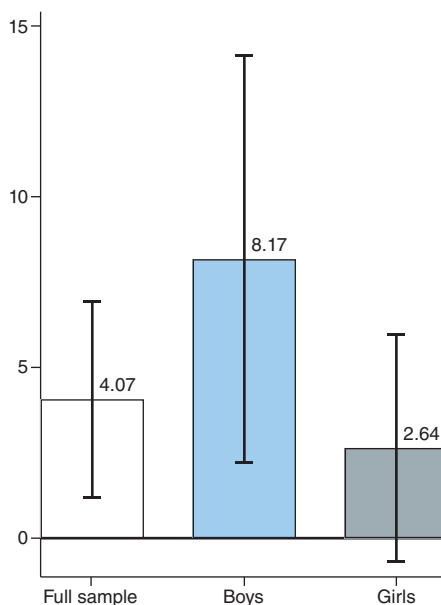


FIGURE 9. EFFECT OF PENSION ON CHILDREN'S WEIGHT-FOR-HEIGHT PERCENTILE IN KIN VERSUS AGE SET SOCIETIES

Notes: Each bar reports the differential effect of pension exposure on households from kin-based versus age-based societies ($\beta_1 - \beta_2$ from equation 5). Moving from left to right, the sample in each specification includes (i) all children under six, (ii) male children under six, and (iii) female children under six. The dependent variable is the child's weight-for-height percentile. Standard errors are clustered by district and 95 percent confidence intervals are reported.

and reduces the likelihood that a child is malnourished by roughly 5.5 percent.³³ The large, positive effect of the pension program on child nutrition in kin-based societies in our sample is completely absent in age set societies.

In the remaining columns, we add an additional set of controls and fixed effects; the estimates remain very similar. In column 2, we include potential pension exposure fixed effects interacted with the age set indicator, thus flexibly absorbing any differences between societies with and without age sets in potential pension exposure and household composition. In column 3, we add age-by-gender fixed effects. In both cases, the estimates are similar and if anything, increase in magnitude. In columns 4 and 5, we investigate the effect for male and female children separately. While we find qualitatively similar results for both, the magnitude is larger for male children, consistent with existing evidence that a disproportionate share of household resources is often invested in boys (Duflo 2003; Milazzo 2014; Barcellos, Carvalho, and Lleras-Muney 2014). Our estimates of $\beta_2 - \beta_1$ from the specifications in columns 3–5 are reported in Figure 9, and are statistically significant ($p < 0.01$) for the full and all-male samples.

³³The estimated effect sizes for kin-based societies are comparable to other estimates from the literature. Duflo (2003) finds that South Africa's pension program increased weight-for-height of girls by 1.19 standard deviations. The pension program she analyzed entailed monthly transfers which corresponded to approximately 248 percent of average income, and she estimated the effect of receiving approximately one year of pensions. If effects on weight-for-height are linear, then our results correspond almost exactly to her findings.

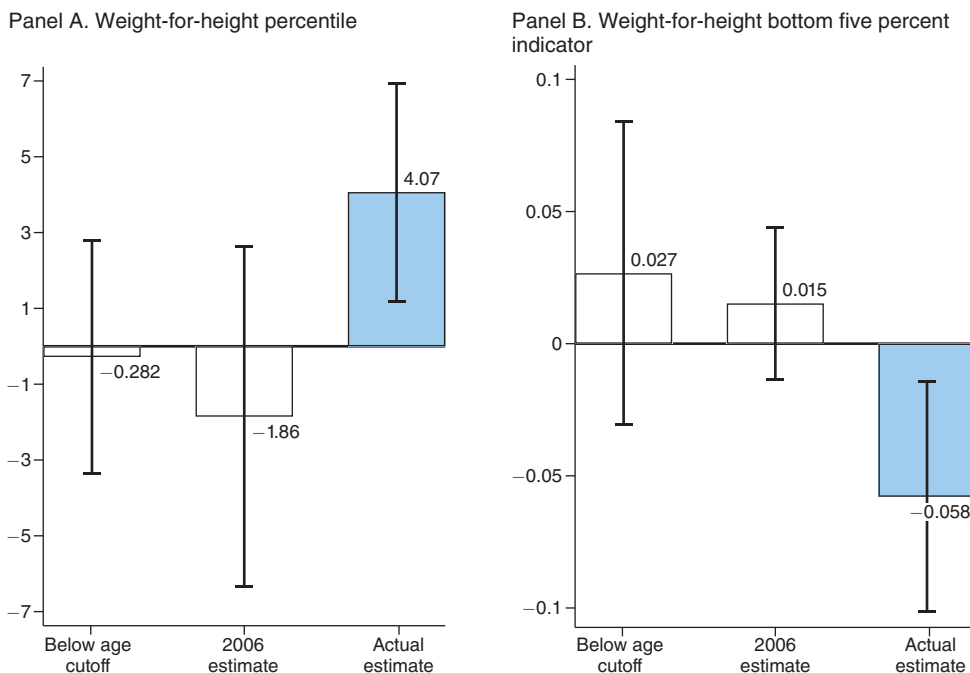


FIGURE 10. FALSIFICATION TESTS

Notes: The left bar of each panel reports the differential effect of *having members aged 55–65* on households from societies without versus with age sets. The center bar of each panel reports the differential effect of *pension exposure measured in 2006* on households from societies without versus with age sets ($\beta_1 - \beta_2$). For comparison, the right bar of each panel reports the differential effect of *actual pension exposure* on households from societies without versus with age sets. In panel A the dependent variable is the child weight-for-height percentile and in panel B it is an indicator if the child falls in the bottom 5 percent of the distribution. Standard errors are clustered by district and 95 percent confidence intervals are reported.

Falsification Tests: To make sure that our estimates capture the causal effect of the pension program, we report two sets of falsification tests, reported in Figure 10. In panel A of Figure 10, the dependent variable is the child’s weight for height percentile, and in panel B of Figure 10 it is an indicator if the child is in the bottom 5 percent of the distribution.

First, we investigate the effect of having household members who are *just below* the age cutoff for the pension program on child nutrition. Any unobservables that lead certain households to have more older members—and any difference between age set and kin-based societies in the inclusion of older members in the household—would likely also affect the presence of older individuals just below the cut off. However, since these individuals do not receive the pension, if our estimates capture the causal effect of the pension program their presence should have no impact on child nutrition. We compute this placebo measure of exposure to household members just below the cutoff following equation (4), except rather than count household members over age 65 in year t , we count household members who would have been aged 55–64 in year t . Our estimate of the differential effect of this placebo exposure measure on child nutrition in societies without versus with age sets ($\beta_2 - \beta_1$) is reported in the leftmost bars of panels A and B of Figure 10; in both cases, we find

no significant effect of our placebo measure and, if anything, the sign is the opposite of our baseline estimates.

Second, we show that our results are not driven by preexisting trends—the patterns we observe are entirely absent prior to the introduction of the pension program. In particular, we estimate equation (5), defining pension exposure exactly as in equation (4) but using data from the 2006 round of the DHS, prior to the introduction of the pension pilot.³⁴ The differential effect of pension exposure in 2006 on child nutrition in societies without versus with age sets ($\beta_2 - \beta_1$) is reported in the middle bar of panels A and B of Figure 10. In both cases we find no significant effect of this second placebo measure and again, if anything, the sign is the opposite of our baseline estimates.

Finally, for comparison, the rightmost bar of panels A and B of Figure 10 reports our baseline (actual) estimates of $\beta_1 - \beta_2$.

Sensitivity Analysis: Measurement and Controls: We report a range of additional estimates that probe the sensitivity and robustness of our baseline findings. First, we show that the findings are very similar using alternative measures of potential pension exposure. The first alternative measure estimates pension exposure only during the survey year, and does not use the household age distribution to retroactively estimate pension receipt in prior years, and the second exploits variation in child age to estimate a more precise proxy for child level exposure. These estimates are described in online Appendix Section B.2 and reported in Table A13.

Second, we document that the results are very similar after including a broad range of controls. We document that the results are robust to controlling for several proxies for household composition and household wealth, as well as interview month fixed effects. We also show that the findings are similar after controlling for several ethnicity-level characteristics; to account for an ethnicity-level characteristic Z_e , we include controls of the form $PotExp_h \cdot \mathbf{1}_d^{Pilot} \cdot Z_e$ along with all single and double interactions. These estimates are described in greater detail in online Appendix Section B.3.

Sensitivity Analysis: Inference: Our estimates exploit differences in social structure across ethnic groups in Uganda, and our main sample consists of 25 groups. Given the relatively small number of groups, we also conduct inference using resampling procedures. As in the analysis of Kenya's HSNP, we generate placebo categorizations randomly assigning each group to either the age-based or kin-based group, both with and without replacement. We compute our estimates of $\beta_1 - \beta_2$ for each placebo assignment and estimate the p -value as the share of placebo estimates larger than our baseline estimates. The full distributions of placebo coefficients, as well as our actual estimates marked with a dotted line, are reported in online Appendix Figure A10. Using the bootstrap, we estimate that $p = 0.024$ for our baseline estimate (column 1 of Table 4), $p = 0.024$ when we include the full set of

³⁴One downside to this analysis using the 2006 round of the DHS is that this round does not ask household members for their ethnicity; instead, it asks for the language they speak at home. While it is straightforward to link languages to ethnic groups, the language information is more aggregated and so we are left with only six languages of which one corresponds to societies with age sets. This feature of the data could also explain the relative imprecision of these placebo estimates.

controls (column 3 of Table 4), and $p = 0.072$ when we restrict the sample to boys (column 4 of Table 4). Using randomization inference, the p -values are $p = 0.000$, $p = 0.000$, and $p = 0.107$, respectively; in the first two cases, none of the placebo estimates are larger than our actual estimate.

Alternative Strategy: Double-Difference Estimates: We next show that the results are qualitatively similar if we *do not exploit variation in household-level pension exposure at all*. A potential shortcoming of our baseline specification (equation (5)) is that it could be biased if the introduction of the pension program induces differential endogenous changes in household composition between members of societies with and without age sets. There are several reasons why this form of selection should not be a major concern, described in online Appendix Section B.4. Nevertheless, to investigate whether our findings are driven by our measured variation in household composition, we present estimates from an alternative specification that only exploits district and ethnicity-level variation, and hence does not depend on household composition at all. These double-difference estimates simply capture the average difference in health outcomes between children from age set versus kin-based societies, in pilot versus non-pilot districts. The details of the estimation are described in online Appendix Section B.4, and the main results are reported in online Appendix Figure A9. Consistent with our main findings, children in kin-based societies (compared to age set societies) are higher on average in the weight-for-height distribution and relatively less likely to be malnourished where the pension pilot was introduced.

E. Additional Outcome Variables

Alternative Measures of Child Nutrition.—In online Appendix Table A15, we investigate the impact of the pension program on a range of additional anthropometric measures. For comparison, column 1 reports the specification in column 3 of Table 4. First, we document that the result is very similar using the weight-for-height z -score (instead of percentile) as the dependent variable (column 2). Second, we show similar results when the dependent variable is an indicator that equals one if the child is below the fifth percentile in the weight-for-height distribution, a common cutoff used by the Centers for Disease Control (CDC) to define malnourishment (Mei et al. 2008; Phillips and Shulman 2018). Age set organization affects not only the impact of the pension program on where a given child falls in the full health distribution but also its impact on the probability that they are extremely malnourished. Third, we document that all findings presented thus far are robust using the weight-for-age distribution rather than the weight-for-height distribution (columns 4–6).

Fourth, we show that the results are qualitatively similar but smaller in magnitude and statistically imprecise using the *height*-for-age distribution (columns 7–9). Measurable changes in child height are only affected by persistent and prolonged patterns of consumption (Hoddinott et al. 2013; Jayachandran and Pande 2017); the average number of years of pension exposure in our sample is 1.9 making it unlikely that the pension program would affect child height by the year of

data collection.³⁵ While the pension program may not be prolonged enough to affect the older children in our sample, one possibility is that young children, who have lived most of their lives since the introduction of the program, are nevertheless affected. To investigate this possibility, we include an additional interaction between our main independent variables and an indicator that equals one if the child is below two years of age, roughly the average years of pension exposure in the sample. These estimates are reported in online Appendix Table A16 and indicate that the pension program does seem to affect the height of the youngest children in kin-based societies, while again having no effect in age set societies.

Finally, we construct an index incorporating all nine measures of child nutrition by computing their first principal component.³⁶ Using this index as the dependent variable, we find a strong, positive effect of pension exposure in societies without age sets and no effect on societies with age sets (difference p -value = 0.0137).

School Attendance.—We finally investigate the impact of pension exposure on the school attendance of primary school age children. While the DHS does not report spending on education, it does report whether school age children actually attend school or not. This is a far cruder proxy for investment in children than either spending on education or child nutrition, since it only captures an extreme extensive-margin decision about whether the child should attend school or not. Nevertheless, Figure 11 reports the differential effect of pension exposure on the primary school attendance of children from kin-based versus age-based societies ($\beta_1 - \beta_2$). The findings mirror our results on the nutrition of young children, except here the findings are strongly driven by male children (middle bar). An additional year of pension receipt increases the likelihood that a primary school age boy is in school by roughly 3 percent in kin-based societies, compared to age-based societies.

Online Appendix Table A18 reports the full set of regression estimates and shows that the effect is driven by a positive and significant effect of pension exposure on school attendance of boys in societies without age sets, and zero effect in age set societies. However, pension receipt does not affect the school attendance of girls in either set of ethnic groups. Given that in our sample most of the ethnic groups are patrilocal, and none are matrilineal, this finding is consistent with Bau (2021) who, in Ghana and Indonesia, finds that grandparents tend to invest in the education of children that will co-reside with them as adults. Online Appendix B.5 further investigates effects on older children, aged 15–18, focusing on both secondary school attendance and teenage marriage.

IV. Discussion: How Culture Shapes Inequality and Vulnerability

This paper's findings show that local social structure shapes financial networks. Income shocks and public policy propagate through the local economy in vastly

³⁵ Our findings are consistent with Cahyadi et al. (2018), who find no effect of cash transfers on stunting after two years of a conditional cash transfer program, where stunting is defined as being two standard deviations less than the WHO's height-for-age standard. Indeed, most analyses of cash transfer programs find no evidence of an impact on child height, especially in the short term (Baird, McIntosh, and Özler 2019; Cahyadi et al. 2018).

³⁶ Intuitively, the first principal component loads positively on all percentile and z -score measures and negatively on all indicators for being in the bottom 5 percent of the distribution (see online Appendix Table A17).

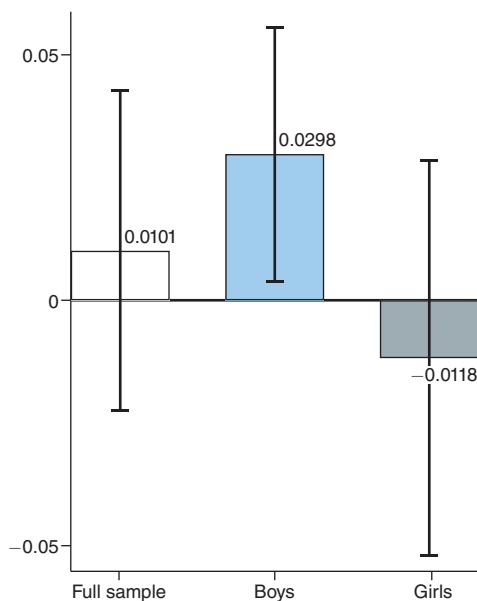


FIGURE 11. EFFECT OF PENSION ON PRIMARY SCHOOL ATTENDANCE IN KIN VERSUS AGE SET SOCIETIES

Notes: Each bar reports the differential effect of pension exposure on households from kin-based versus age-based societies ($\beta_1 - \beta_2$ from equation 5). Moving from left to right, the sample in each specification includes (i) all children 6–14, (ii) male children 6–14, and (iii) female children 6–14. Standard errors are clustered by district. Coefficient estimates and 95 percent confidence intervals are reported.

different ways across ethnic groups. The distinct set of economic ties that exist in kin-based compared to age set societies suggests that social structure could also be an important determinant of the pattern of cross-sectional inequality. It could also imply that the two forms of social organization are well positioned to weather very different types of economic shocks and overcome distinct economic challenges.

In age set societies, we find that financial ties are strongest within the age cohort. Since individuals rely disproportionately on their own age cohort for economic support, people at low-earning parts of the life cycle could be left systematically worse off. The young and the old, for example, must rely (largely) on other members of their own age cohort, who have similarly low income-earning potential. In kin-based societies, however, where individuals are likely to have strong connections to people at all stages of life, inequality over the life cycle could be lower. For example, older individuals can rely on younger generations and young people can rely on older generations.

A very different pattern, however, may exist across family lineages or clans. In kin-based societies, where resources are kept to a larger extent within the extended family, certain clans may accumulate substantially more wealth and opportunity than others. Positive and negative shocks alike propagate through the extended family. In age set societies, where age sets cut across lineage, this form of inequality could be more limited since the age set system links individuals in different families.

To investigate this set of hypotheses about the structure of inequality, we return to our baseline consumption data from the HSNP survey in Kenya (Oxford Policy

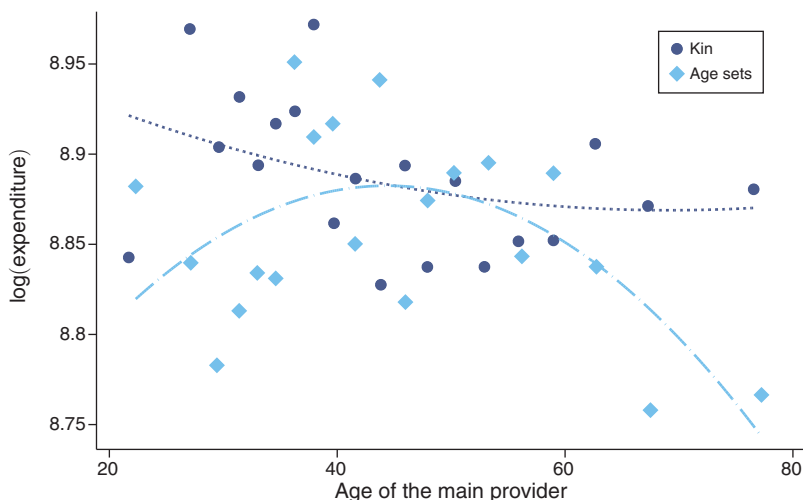


FIGURE 12. CONSUMPTION OVER THE LIFE CYCLE: AGE SET VERSUS KIN

Notes: Binscatter plots of log of household consumption expenditure (y-axis) versus the age of the main provider (x-axis), separately for members of age set societies (light diamonds) and kin-based societies (dark circles). Quadratic fit curves are also displayed. We partial out sublocation fixed effects and the number of household members.

Management 2010). Figure 12 reports total consumption expenditure for male main providers across age cohorts, separately for societies with and without age sets. Consistent with our conjecture, old and young members of age set societies consume systematically less, while the evolution of consumption over the life cycle in kin-based societies is comparatively flat. This difference in consumption patterns is statistically significant, as documented in online Appendix Table A20, and does not seem to be driven by differences in *income* over the life cycle in societies with and without age sets.³⁷ The combination of this finding with the main results of the paper indicates that the preeminence of financial ties within the age cohort in age set societies generates less consumption smoothing over the life cycle and relatively less consumption on average in old and young ages.

Our hypotheses also suggest that while there is greater inequality across age cohorts in age set societies, there could be systematically *less* inequality across extended families or clans. Figure 13 reports the average consumption spending of all households in each sub-clan, separately for members of age-based and kin-based societies. Sub-clans are ordered within the age set and kin-based groupings based on the average level of consumption. Consistent with our conjecture, the figure shows substantially less consumption inequality across sub-clans in age set societies. The pattern for age set societies is relatively flat, while in kin-based societies there are large consumption differences across clans, particularly those at the far left and far

³⁷ We estimate the relationship between (log of) income and age and age², separately for kin-based and age-based societies. We find no evidence of a heterogeneous relationship between age and earned income for societies with versus without age sets. The coefficients on the interaction terms between the age set indicator and age and age² respectively are $-0.00314(0.00378)$ and $0.00001(0.00006)$.

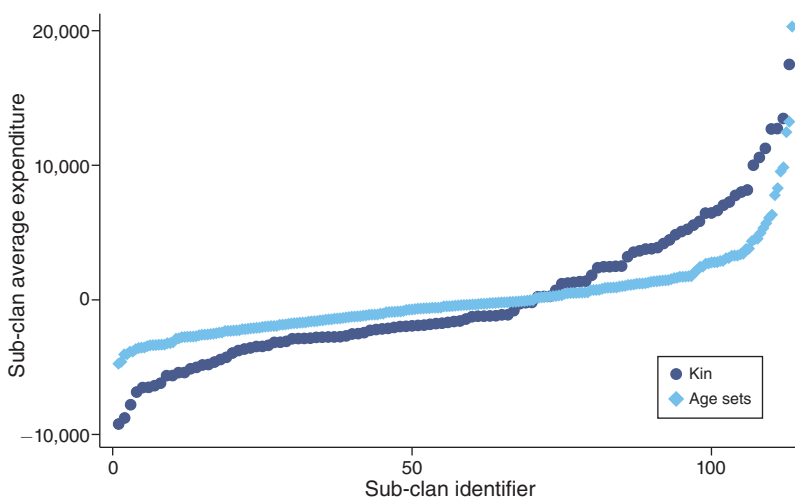


FIGURE 13. CONSUMPTION ACROSS SUB-CLANS: AGE SET VERSUS KIN

Notes: Scatter plot of consumption expenditure (y-axis) versus sub-clan identifiers (x-axis) plotted separately for members of age set societies (light diamonds) and kin-based societies (dark circles). Sub-clans are ordered by their average consumption spending, which is normalized by the mean consumption spending for each series.

right ends of the figure. To put the magnitude in perspective, the standard deviation of consumption across sub-clans is 60 percent larger in kin-based societies compared to age set societies; the interquartile range (75–25) of average consumption across sub-clans is KSh 2870 in age set societies and KSh 5594 in kin-based societies, corresponding to 56 percent of the in-sample average annual income.

Together, these findings indicate that the two forms of social organization lead to very different patterns of inequality and are likely well positioned to withstand different forms of hardship. Shocks or forms of economic hardship that affect specific age cohorts could be more challenging to withstand in age set societies. These may include conflict (which affects predominately young men); diseases that disproportionately affect specific cohorts (HIV in the case of the young and COVID in the case of the old); as well as the fact that, as discussed above, the young and the old have less access on average to income earning opportunities. On the other hand, shocks to specific locations or lineages could be harder to weather in kin-based societies. These may include the bankruptcy of a family firm or farm; adverse weather conditions in a particular region; divorce and lineage-based feuding; or the out-migration of younger members of the group, leaving few people to take care of older members of the family. The dynamic impacts on wealth accumulation of these two forms of network structure could also be very different. Inequality across clans can grow over time, which might have adverse consequences, including the accumulation of political inequality and coercion of poorer sub-clans. The lower bequest motives in age set societies, however, as evidenced by the absence of investment in children in either of the two pension programs that we analyze, could lead to less intergenerational accumulation across the board. We leave these implications of social organization to future work.

Methodologically, our results suggest that a deeper understanding of social structure, and anthropological analyses thereof, could be a useful complement to direct network measurement by making it possible to proxy the structure of social networks in contexts where direct measurement is impossible or prohibitively costly. This is especially important in light of mounting evidence that network structures play a key role in analysis of risk sharing and development policy (Breza et al. 2019). Using our reading of existing ethnographic work to identify age set organization, we were able to predict where particular social ties would be strong and weak, and hence how both Kenya's cash transfer program and Uganda's pension program would propagate through the economy. We were also able to predict which segments of the population would be most vulnerable on average. The approximation that our method provides could be useful for policy analysis and design, especially for large-scale programs or in more remote areas where network data collection is not an option.

V. Conclusion

This paper documents that social structure predictably shapes patterns of economic interaction, informal redistribution, and the impact of public policy. A rich ethnographic literature suggests that local differences in social structure, and in particular the distinction between age-based and kin-based social organization, are important for financial transfers. In age set societies, individuals are organized into age-based social groups and an individual's relationships with his "age mates" often take priority over family ties. Yet despite the prevalence of age set organization in sub-Saharan Africa and around the world, its economic implications have not been studied quantitatively.

To investigate these ideas, we collect information from a broad range of ethnographic sources on the social structure of ethnic groups comprising the majority of the population of Kenya and Uganda. Exploiting unique features of a cash transfer experiment in Kenya, we first document that in age set societies there is strong evidence of within-cohort financial ties and zero evidence of within-clan financial ties or intergenerational investment within the family. In kin-based societies we observe the opposite pattern. Second, analyzing the introduction of Uganda's national pension program, we show that pension grants have large, positive spillover effects on child nutrition in kin-based societies, but no effect in age set societies, where financial ties across generations are weaker. Local social structure leads to strikingly different distributional consequences of public policy, suggesting that insights from anthropology about variation in local financial networks could inform economic and policy analysis.

Age set organization is one example of the vast differences in social structure that exist around the world, and of how patterns of allegiance and economic exchange may vary widely across contexts. Other examples include secret societies, cults and religious groups, gender-based groups, military associations, and clan super-sets that link individuals across broad geographic expanses. While our analysis focuses specifically on age sets, our findings suggest that existing descriptions of social organization could be important for better understanding and predicting differences across contexts in patterns of economic exchange. Exploring additional features of social organization

and how they shape modern financial networks, inequality, and the heterogeneous effects of public policy strikes us as an exciting area for future work.

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